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**“Alternative Investments in the Context of Portfolio Diversification:
Implications for major institutional U.S. investors”**

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Abstract

The paper is an overview of alternative investment vehicles typically chosen for diversification purposes by major U.S. institutional investors with long term investment horizon. The study determines the state and diversification qualities of various alternative asset classes during 29 years (from 1997 to 2025 inclusively) on a selection of net-of-fees total return alternative investment indices employing econometric methods accounting for appraisal smoothing and lagged economic relationships to develop an arbitrage pricing theory (APT) style parsimonious framework for estimating alternative instruments risk exposure. Some notable results of the analysis include confirmation of Alpha existence for gold and optimum yield commodity futures while proving Alphas to become insignificant for hedge funds, real estate and Bitcoin after accounting for appraisal smoothing and structural breaks.

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Introduction

Major institutional investors are constantly assessing ways to preserve long-term portfolios through diversification. Alternative assets are increasingly becoming a staple for such portfolios thanks to a variety of diversification benefits they bring, such as high returns (private equity), inflation hedge (real assets), asymmetric bets (venture capital) and low correlation to traditional assets (commodities).

Entities historically averaging substantial alternative investments are pension funds, university endowments and insurance companies. For the aforementioned entities, hedging through diversification (in a vein akin to immunisation, implied herein as liability-driven investing) is a question of paramount importance: long-term liabilities, typically carried by each institution, create a considerable necessity to maintain positions which could survive changing cycles and market crashes, maintaining the institution's solvency.

Following the success of Yale Model, emphasising aggressive diversification into illiquid assets, pioneered by David Swensen and Dean Takahashi, university endowments allocations in alternative assets grew from 27% in 2003 to 56% in 2023 ([PE 150, 2025](#)). Digital assets have recently been growing in popularity as risk-on assets with amplified returns; however, the allocation is considered high-risk and is usually limited to 1-5% of the portfolio (if any)¹.

Current macroeconomic and market conditions exhibit developments which make traditional portfolios more exposed and force the investors to reconsider their diversification choices, making it crucial to re-evaluate the empirical meaning and constitution of a diversified portfolio. Molenaar et al. ([2024](#)) report increasingly positive correlation between U.S. stocks and bonds,

¹ Citation needed.

and increasing volatility of traditional 60/40 portfolios. High inflation and high volatility regime further complicate preservation of wealth. As of June 2026, the Magnificent Seven amounted to approximately \$23 trillion and 35% of total U.S. market cap, creating unprecedented equity concentration risk. According to World Gold Council ([2025](#)), the central banks have purchased over 1000t of gold each year from 2022 through 2024. The aforementioned conditions create a strong demand for alternative investment vehicles, providing risk profiles different from traditional stocks and bonds.

The diversification benefits of alternatives have long remained a topic for debates, creating opposing camps and enabling views ranging from highly positive to negative, with notable works, such as Platanakis et al. ([2019](#)) and Phalippou ([2020](#)), arguing against some of the mainstream assumptions of the persistence of private capital outperformance of the public market equivalents and even against the actual reporting factuality, justifiably claiming measurement errors and numerous biases inherent to private capital reporting and appraisal methodology. Some of the biases are discussed during the course of the research.

The methods to account for aforementioned biases are constantly evolving together with the asset classes themselves. Changing market regimes and emergence of new asset classes and strategies make a comprehensive overview and exception daunting task, which is, however, partaken in the research.

The major idea, however, is to continue the arbitrage pricing theory tradition, following Ross, Ilmanen and Ang, to promote a meaningful decomposition of the various asset classes returns into associated risk premia, which could allow a more nuanced approach to asset allocation than a simple mean-variance optimisation would allow. This aligns well with the long-term planning nature of the major institutional investors, enabling scenario analysis and a considerably more flexible approach to estimating the relative place of the instruments in the portfolio with regards to underlying risk exposures.

The formal objectives of the present study are: collecting a panel dataframe of quarterly net-of-fees total return indices representing alternative asset classes; correcting smoothing biases present in hedge funds, private capital and private real estate indices; assessing asset classes statistics and correlation; estimating risk exposure and abnormal return using a factor-based framework. The research hypotheses are: the existence of genuine Alpha for gold and commodities; insignificance of Alpha for hedge funds, private capital and real estate after unsmoothing; existence of crisis Alpha for managed futures; ability of the traded factors framework to span most of the asset classes in the research.

The paper continues according to the following structure: in [Literature review](#) section, the preceding works are analysed and the relative standing of the paper is defined; [Definitions](#) section serves to identify the scope of the research by providing exact definitions of the research objects; [Data description](#) lists the indices used to represent the asset classes and the macro factors, and provides the rationale behind the indices choice; [Research design](#) defines the research methods layout and sequence; [Findings](#) section gives an overview of the results; [Discussion](#) section elaborates on the results; [Conclusion](#) section gives a summary of the insights acquired; and, finally, the [Limitations and further research](#) section lists limitations, activities slightly beyond the scope of the paper and provides directions for continued development.

Literature review

Research on diversification benefits of the alternative assets appears to be a relatively young topic by the standards of the academia. Amin & Kat ([2003](#)) investigated claims of hedge funds providing superior risk-return trade-offs and demonstrated hedge funds, while providing no true enhanced returns on their own, proved to be an efficient diversifier when mixed with S&P 500. The first major work analysing inclusion of a varied collection of alternative assets into the portfolio ([Fischer & Lind-Braucher, 2009](#)) focused primarily on optimising portfolios accounting for varying statistical metrics, such as volatility, value at risk (VaR), conditional value at risk (CVaR) and their modified variants, accounting for higher moments (skewness and kurtosis), as well trying different estimates of expected returns. The majority of papers until recently continued in a similar vein, examining statistical properties and mostly applying mean-variance optimisation. Platanakis et al. ([2019](#)) approached estimation of harm from alternative assets diversification and argued that diversification can be harmful to U.S. investors primarily due to estimation errors, amplifying excessively during crises, raising major concerns on biased estimations. Fung & Hsieh ([1997](#)) first started applying factor models in the style of Fama-French to explain performance of hedge funds, paving way to further factor-based frameworks.

A large part of the alternative asset classes universe consists of different forms of private assets, subject to a multitude of biases, such as selection bias, appraisal smoothing, lagged reporting etc. A method to unsmooth appraisal-based returns has been proposed by D. M. Geltner ([1991](#)) and polished in subsequent works (D. M. Geltner ([1993](#)), Clayton et al. ([2001](#)), D. Geltner et al. ([2003](#))) primarily with application to real estate. Aside from real estate, the method proved useful for unsmoothing other types of private capital appraisals. Getmansky et al. ([2004](#)) proposed a model addressing the nature of smoothing and liquidity

exposure in hedge funds returns. Coutts et al. ([2023](#)) developed a three-step unsmoothing procedure of fund-level returns to account for undercorrection and residual correlation remaining after the Geltner and Getmansky-Lo-Makarov application (which is unfortunately inapplicable in the present study due to data limitations). Each of the methods (especially the latter) is expected to significantly increase risk exposure of the unsmoothed series and reveal the true economic pattern. Liang ([2000](#)) finds hedge funds survivorship bias can exceed 2% per year. Coming from a previous research ([Degeorge et al., 2016](#)) demonstrating underperformance of secondary buyouts made under pressure to spend, Phalippou ([2020](#)) investigates private capital funds' fees structure and applies multiplier of money (MoM) and public market equivalent (PME) calculations to reveal that private equity produced no Alpha for investors since 2006, performing largely on par with small-cap indices. The approach, however, cannot be replicated in the present study due to data limitations (lack of detailed fund-level data). Brown et al. ([2024](#)) apply stochastic discount factor (SDF) and portfolio alpha to provide an argument on the contrary, showing the buyouts' cash flows to move differently from the public markets following different underlying macro factors and estimating buyouts' Alpha to be statistically significant on a period from 1987 to 2022 for a large sample of private funds. Asness ([2026](#)) argues the increasingly positive correlation between stocks and treasury bonds, pushing investors towards alternative assets, is an insufficient reason to diversify into arbitrary alternative asset classes, since many of them are exposed to equity risk.

The research is heavily inspired by arbitrage pricing theory ([Ross, 1976](#)). Chen et al. ([1986](#)) demonstrates that macroeconomic variables systematically affect stock market returns. Ilmanen ([2012](#)) refines the approach into a comprehensive framework, emphasising forward-looking indicators. Ang ([2014](#)) goes even further, positing the asset classes are merely repackaged underlying risk factors, decomposing each asset into a sum of its risk premiums, advocating for systematic approach and a focus on market downturns. Pedersen et al. ([2014](#))

attempt to develop a unified risk-factor framework for alternative assets, notably using transformed risk factor returns and lagged betas - admittedly a concept close to what this paper tries to achieve.

When considering portfolios of alternative assets, prior literature mostly focused either on mean-variance style optimisation techniques or on studying single asset classes in isolation, which greatly limited the overall scope. Hereby, it is argued, that statistical properties and correlations between the asset classes alone are insufficient to capture the true diversification benefit of alternative instruments (or lack thereof), which is the main premise of the present study. The research aims to construct a parsimonious APT-style framework for alternative asset pricing, consisting of traded factors, which would balance efficiency and simplicity of interpretation, being easily applicable for prototyping risk exposure of major U.S. investors allocations. It is also aiming to account for the smoothing and lagged economic relationships to the best extent possible without accessing proprietary fund-level data, cover a broad asset classes universe and a considerable time span of 29 years of quarterly returns. Besides, the study speaks to the common debates on persistence and significance of various results, routinely raised in the aforementioned works, such as the existence of “crisis Alpha” for managed futures, the reality vs. illusion of private capital performance, diversification benefits of real estate and digital assets, and whether each result holds after accounting for smoothing and lagged economic relationship.

Definitions

A “major U.S. institutional investor” in the context of the research largely corresponds to the definition under SEC Rule 15a-6 defined by Proskauer Rose LLP ([Castellon et al., 2020](#)):

“‘Major U.S. institutional investor’ means a U.S. institutional investor with assets, or assets under management, in excess of US\$100 million, or a registered investment adviser with assets under management in excess of US\$100 million.”

Alternative assets are loosely defined as follows by Mundi & Kumar ([2022](#)):

“Alternative investments do not have a uniform definition and include hedge funds, private equity, real assets and structured products in the broadest categories of alternative investments (Chambers et al., 2020). As evident in the present research, cryptocurrencies, structured products, art work, initial public offers (IPOs) and venture capital are also part of alternative investments.”

The present study introduces a more precise list of alternative assets. The assets are aggregated specifically to reflect major U.S. institutional investors’ portfolios with diversification properties in mind:

Traditional assets include *public equities, corporate investment grade bonds, corporate high yield bonds, government bonds* and cash and cash equivalents (the latter not represented in the paper due to focus on long investment horizon).

Alternative assets include private capital (namely *buyout, venture capital* and *private debt*), real assets (namely *private real estate*, and *commodities*, utilising optimum yield futures strategies to avoid negative roll yield), *gold* (specifically separated from commodities for its monetary properties), *hedge*

funds (including a specific subsample of *managed futures* industry), digital assets (represented by *Bitcoin*).

Omitted from the study are *private infrastructure* (due to unavailability of data) and *collectibles and niche* such as art, classic cars, and wine (due to not being a meaningful allocation for major U.S. institutional investors).

Data description

General considerations

Each asset class in the research is represented by a specific index chosen on the criteria of availability, historical depth (spanning from 1997 to 2025 with the exception of Bitcoin), acceptance by both academia and practitioners (major U.S. institutional investors specifically) and representative quality for the specific asset with a focus on diversification potential. Traditional assets are included alongside alternative to provide a clear benchmark and comparison.

Every index is chosen to be a form of total return as opposed to net total return (with the only exception of gold spot prices, discussed in [Gold](#) section). The reason behind such choice are tax exemptions for many of the entities (such as pension funds and endowments) classifying as major U.S. institutional investors, elimination of arbitrary theoretical tax assumptions, and accounting for a manager's capacity to capture the full market return before tax. While not necessarily being the correct measure of the actual return in every case, total returns provide a clearer unobscured comparison than net total.

Each index including fund performance or transaction data (namely, the indices for hedge funds and managed futures, private capital and private real estate) is net-of-fees. Eliminating fees is required due to their arbitrary nature, complex non-uniform structure and substantial amounts. The rest of the indices (namely, public equities, bonds, gold, Bitcoin and optimum yield commodity futures strategies) do not account for fees or transaction costs. However, in case of those instruments, in the long term the costs are negligible, making the indices comparable.

Every index is then transformed (aggregated and, if needed, unsmoothed²) to be comparable within a single panel dataframe. The entire further analysis

² See [Unsmoothing non-traded indices](#) section further.

operates with geometrically compounded quarterly excess returns calculated as total returns (spot returns for gold) minus risk-free rate to isolate true Alphas, Betas and correlation.

The paper is specifically focusing on U.S. market and investors with long investment horizon loosely defined as 10 years, hence the choice of indices and asset classes is specifically constrained to the aforementioned domain. It's worth acknowledging that major U.S. institutional investors invest in assets globally, however, that is outside the scope of the paper.

Traditional instruments

Equities

Equities are represented by S&P 500 Total Return Index Value, a float-adjusted market capitalisation weighted index including 500 leading U.S. companies with large market capitalisation, roughly covering 80% of available U.S. market capitalisation. The index is commonly used in academia and business and covers equities which are typically chosen by major U.S. institutional investors, thus preferred over broader indices such as CRSP US Total Market Index or Russell 3000 Index in the research. It should be acknowledged that large scale institutional U.S. investors do hold small and medium capitalisation stocks as well as international equities which go beyond the scope of the paper.

Treasury bonds

Treasury bonds are represented by a custom U.S. aggregated treasury bonds total return series computed from FRED U.S. Treasury Constant-Maturity Yields of different maturities using the duration-convexity approximation following Swinkels ([2019](#)) and static weights per maturity in the style of Bloomberg US Treasury Total Return Unhedged USD (LUATTRUU)³. The primary reason for

³ The exact weights for each maturity series are: $1Y = 0.10$, $2Y = 0.13$, $3Y = 0.10$, $5Y = 0.22$, $7Y = 0.12$, $10Y = 0.16$, $20Y = 0.09$, $30Y = 0.08$

choosing this approach is unavailability of quality data (such as the aforementioned Bloomberg index).

Corporate bonds (investment grade and high yield)

Corporate bonds are represented by two subsamples: investment grade and high yield bonds. This distinction serves to separate two very different asset classes, each having its own specific purpose in a major U.S. institutional investors' portfolio: investment grade bonds are used to match long-term liabilities, whereas high yield bonds provide enhanced yield.

Investment grade bonds are represented by ICE BofA US Corporate Index Total Return Index Value. According to FRED website, the index tracks the performance of US dollar denominated investment grade rated corporate debt publicly issued in the US domestic market.

High yield bonds are represented by ICE BofA US High Yield Index Total Return Index Value. According to FRED website, the index tracks the performance of US dollar denominated below investment grade rated corporate debt publicly issued in the US domestic market.

Alternative instruments

Gold

Despite being a widely used instrument, gold is technically considered an alternative asset. It is often separated from commodities and put into a category of its own for its monetary qualities. Investible gold market capitalisation in U.S. is relatively small (\$6.7 trillion vs \$60.8 for S&P 500 and \$30.27 for treasury bonds as of 31 December 2025) the primary reasons being cultural investment preferences for higher return assets and relatively limited private holdings in comparison to U.S. government reserves. Gold is represented by London Bullion Market Association Gold Price PM Fix, a global benchmark gold price set at 3 PM GMT daily (coinciding with the opening time of major U.S. exchanges) and

used to settle physical transactions. Major U.S. institutional investors typically make long-term allocations to spot gold rather than futures. Calculation of total return is problematic; however, the opportunity cost of financing is accounted for via risk-free rate subtraction and the cost of carry is negligible, hence, spot price is considered sufficient for comparison to the other total return indices in the research.

Digital assets (Bitcoin)

Digital assets, such as Bitcoin and Ethereum, are an increasingly popular investment vehicle for major U.S. institutional investors, serving as a high risk and high return vehicle largely uncorrelated to the rest of the market. The allocations however are considered risky and remain relatively small, at 1-5% of the portfolio maximum⁴ (if any). Being the most prominent instrument, Bitcoin is used to represent the industry in the research. Bitcoin is represented by Coin Metrics Bletchley Index Bitcoin - Total Returns. According to CMBI website, the index is designed to measure the performance an investor would expect from purchasing and holding Bitcoin as well as liquidating any forked assets that meet the criteria defined in the CMBI Fork Legitimacy Policy. CMBIBTC is an increasingly acclaimed benchmark, tracking Bitcoin price as far back as 2010, while total return variant also accounts for the forks, representing, in a way, the “yield” part of the asset and making it comparable to other total return indices covered in the present study. It should be noted, that unlike all other series starting in the analysis going as far back as 1997, Bitcoin series only starts in 2010 complicating direct comparison of statistical properties.

Commodities (optimum yield commodity futures strategies)

Commodities are typically held by major U.S. institutional investors as diversified futures baskets. Commodities futures often suffer from negative roll yield, occurring when the market is in contango, where long-term contracts are

⁴ Citation needed.

more expensive than short-term, forcing investors to sell expiring contracts cheaper and buy more expensive. Consequently, major U.S. institutional investors don't simply passively hold the futures, but invest in exchange-traded funds (ETF) implementing active strategies, such as optimum yield approach (Deutsche Bank), select roll and enhanced roll (Bloomberg) and dynamic roll (S&P). In the present study, commodities are represented by Deutsche Bank Index Quant Optimum Yield Diversified Commodity Index Total Return, which is the index tracked by Invesco Optimum Yield Diversified Commodity Strategy No K-1 ETF, one of the largest and most popular commodities ETFs, used to hedge against inflation and diversify the portfolio. According to DBIQ Optimum Yield Diversified Commodity Index Guide 2024:

“DBIQ Optimum Yield Diversified Commodity Index is based on 14 commodities drawn from the energy, precious metals, industrial metals and agriculture sectors. ... Deutsche Bank Liquid Commodities Indices Optimum Yield (DBLCIOY) are designed to maximize potential roll returns by selecting, for each commodity, the futures contract with the highest implied roll yield. The index aims to maximize the potential roll benefits in backwardated markets and minimize the loss from rolling down the curve in contango markets.”

Hedge funds

Hedge funds are represented by Barclay Hedge Fund Index. According to BarclayHedge website, the Barclay Hedge Fund Index is a measure of the average return of all hedge funds (excepting Funds of Funds) in the Barclay database. The index is simply the arithmetic average of the net returns of all the funds that have reported that month. It should be acknowledged, that major U.S. institutional investors typically seek exposure to the better performing funds, making the broad index a suboptimal choice, however, due to data limitations, the broad index is used in the present study.

Managed futures

Managed futures, often employed by commodity trading advisers (CTA), are strategies aiming to trade futures across commodities, currencies and financial markets. The strategies are designed to make profit independently of the market direction, and being uncorrelated to public markets, serve as a primary diversification factor. While technically being a subset of the broader hedge funds collection covered in [Hedge funds](#) section, such funds are specifically favoured by major U.S. institutional investors as diversification vehicles, and exhibit properties vastly different from the overall umbrella of various funds, and therefore are included as a separate asset class in the research. Managed futures are represented by Barclay BTOP50 Index. The index is described on BarclayHedge website as follows:

“The BTOP50 Index seeks to replicate the overall composition of the managed futures industry with regard to trading style and overall market exposure. The BTOP50 employs a top-down approach in selecting its constituents. The largest investable trading advisor programs, as measured by assets under management, are selected for inclusion in the BTOP50.”

Unlike the broader Barclay Hedge Fund Index, BTOP50 does not exhibit autocorrelation, hence does not require unsmoothing.

Real estate

Private real estate allocation is a popular way of hedging against inflation, stable income from rent, perceived uncorrelatedness to and lower volatility than public markets - the latter notions being argued against in previous researches as well as the present study. Private real estate is represented by National Council of Real Estate Investment Fiduciaries Fund Index - Open End Diversified Core Equity. According to NCREIF, the index is a capitalisation-weighted, time-weighted return index tracking the performance of open-end funds employing

lower risk investment strategies utilizing low leverage and generally represented by equity ownership positions in stable U.S. operating properties diversified across regions and property types. The index is based on quarterly appraisals. In the present paper, net-of-fees data is used.

The private real estate index is benchmarked against its listed *public market equivalent (PME)* counterpart, represented by Financial Times Stock Exchange Nareit U.S. Real Estate Index Equity REITs⁵.

Private capital

Private capital became an increasingly popular alternative investment vehicle after 2008 global financial crisis thanks to high returns and diversifying properties. In the present study, private capital is represented by a set of State Street Private Capital Indices providing time-weighted returns. Being one of the largest custodian banks, State Street leverages its own database of custodial client transactions, eliminating self-reporting bias from the indices. However, it should be acknowledged, that the indices introduce a different selection bias, since all the constituents are State Street clients, which may be a sample different from the general population of clients of all custodian banks. According to State Street website, the State Street Private Capital Indices are predefined benchmark series measuring the aggregate performance of private capital funds across a comprehensive range of asset classes, strategies and geographics. The benchmarks include buyout, venture capital and private debt, each included in the study. Public market equivalents (PME) are provided

Buyout is a private equity strategy aiming to acquire controlling stakes in mature companies to improve profitability over a 5-10 years horizon. Major U.S. institutional investors often allocate significant parts of their portfolios to buyout funds acting as leveraged equity with illiquidity premium for higher returns. It has to be acknowledged, that a proper PME (such as Russell 2000 or Russell 3000)

⁵ See [Unsmoothing non-traded indices](#) section further.

could not be obtained for the required sample period (1997-2025) due to data limitations.

Venture capital is a private equity strategy aiming to acquire stakes in early-stage startups with high growth potential. Major U.S. institutional investors act as limited partners, allocating to venture capital funds primarily for enhanced returns and innovation exposure. PME for venture capital is selected to be NASDAQ Composite Index, featuring a similar statistical profile and a similar strong weighting towards technology, biotech and consumer services.

Private debt is a practice of non-banking entities lending to the companies directly. Major U.S. institutional investors access private debt through commingled funds, business development companies (BDC) and separately managed accounts (SMA) for higher yield and lower correlation to public markets than public bonds. PME for private debt is selected to be Morningstar LSTA US Leveraged Loan Index, featuring a floating rate and seniority, which is inherent to direct lending, which constitutes the vast majority of the SSPCI Private Debt index.

Infrastructure

Finally, a word of acknowledgement has to be made, that a high-quality infrastructure index data could not be found for the study, reducing the scope of the paper in regards to breadth of the research. Infrastructure is another increasingly popular asset class among institutional investors. Exposure is achieved both through public equities and private ownership, the latter being specifically beneficial for diversification via low correlation to public markets, low volatility, stable income, inflation hedge, illiquidity premium and exposure to the underlying type of business activity. Exposure to commodities and energy sector, traditionally done via public stock ownership and trading futures, is also increasingly exercised via private infrastructure ownership. However, due to lack of data, the topic is not covered in the present study.

Macro factors

One of the key sections of the current research is factor investment approach analysis in the spirit of arbitrage pricing theory (APT), loosely following BlackRock factor investing framework ([BlackRock, 2026](#)) and the research of Chen et al. ([1986](#)), Ilmanen ([2012](#)) and Ang ([2014](#)). Each factor is designed to be a tradeable, market-based indicator. The factors are chosen to represent the major systematic risks affecting major U.S. institutional investors' portfolios⁶.

Risk-free rate

Risk-free rate is represented by 1-month treasury bills simple periodic returns (without reinvestment) acquired from Kenneth R. French Data Library. According to [Kenneth R. French's website](#):

“The one-month Treasury bill rate data through May 2024 are from Ibbotson Associates. Starting from June 2024, the one-month Treasury bill rate is from ICE BofA US 1-Month Treasury Bill Index.”

Monthly arithmetic returns are geometrically compounded to acquire quarterly returns for uniform comparability to the rest of the indices.

Market

Broad equity market factor is used to determine the risk premium of owning a diversified set of public stocks. It is represented by Fama-French excess return on the market ($Mkt-R_f$) from Kenneth R. French Data Library. According to [Kenneth R. French's website](#):

“ $R_m - R_f$, the excess return on the market, [is the] value-weight return of all CRSP firms incorporated in the US and listed on the NYSE, AMEX, or NASDAQ that have a CRSP share code of 10 or 11 at the

⁶ More in [APT-style factor framework](#) section further

beginning of month t , good shares and price data at the beginning of t , and good return data for t minus the one-month Treasury bill rate.”

Monthly arithmetic excess returns are geometrically compounded to quarterly.

Commodities

Commodities factor determines whether the instrument is resilient to broad commodities price changes. *Commodities and Term factors combined are expected to explain the major part of inflationary pressure.* In this paper, the price change is represented by quarterly arithmetic returns of [Bloomberg Commodity Index \(BCOM\)](#) price level:

$$r_t^{\text{BCOM}} = \frac{P_t}{P_{t-1}} - 1$$

Term

Term factor measures sensitivity of the assets classes to shifts in the yield curve, or the difference between long and short-term interest rates, as well as monetary policy stance, inflation expectations and overall economic outlook. *Commodities and Term factors combined are expected to explain the major part of inflationary pressure.* In the present study, is calculated as the difference between Swinkels-approximated total return constructed from end-of-quarter yields of [10-year treasury bonds \(DGS10\)](#) from FRED and the risk-free rate (1-month treasury bills) acquired from [Kenneth R. French's website](#).

$$r_t^{\text{term}} = TR^{10Y \text{ T-Bonds}} - TR_t^{1M \text{ T-Bills}}$$

Credit

Credit factor reveals sensitivity to changes in credit spreads. In the research, it is calculated as the difference between [ICE BofA US Corporate Index Total Return Index Value \(BAMLCC0A0CMTRIV\)](#) and a weighted average Swinkels-

approximated total return constructed from end-of-quarter yields of [3-year \(DGS3\)](#) and [5-year \(DGS5\)](#) treasury bonds from FRED:

$$r_t^{\text{credit}} = TR_t^{\text{HY Bonds}} - TR_t^{3-5Y \text{ T-Bonds}}$$

It is a measure of risk premium for holding below investment grade bonds over treasuries of the (roughly) same maturity. High yield is preferred over the corporate equivalent for higher responsiveness and being a much better match for many of the alternative assets in the present study (digital assets, hedge funds, private capital, commodities).

The premium is strongly correlated with equity market ($\rho \approx 0.76$, Table 8), credit premium growing together with the market returns. The two factors thus create considerable multicollinearity (*credit VIF* ≈ 3.49 , Table 9), which could lead to a certain degree of instability of market and (especially) credit coefficients. However, since VIF doesn't cross the commonly accepted threshold of 5, the choice was made to keep the factor as-is without further modifications in order to preserve straightforward economic interpretation.

Liquidity

Liquidity factor determines sensitivity of the assets to ease of funding access. It is particularly important for alternative assets, since many types of alternatives are, by nature, illiquid and depend on availability of credit and well-functioning secondary markets. Liquidity is represented by [Pastor-Stambaugh traded liquidity \(LIQ_V\)](#) ([Pastor & Stambaugh, 2003](#)), a measure of premium investors require for holding illiquid stocks. Quarterly changes are calculated as quarterly geometrically compounded percentage changes of the monthly change given in the Lubos Pastor's dataset.

Currency

Currency factor determines whether the assets are affected by the trade-weighted value of U.S. dollar. The factor influences commodity prices, cross-

border capital flow and international competitiveness. It is represented by the [ICE U.S. Dollar Index](#). According to ICE website:

“The DXY is a geometrically averaged calculation of six currencies weighted against the U.S. dollar, and is calculated intraday using the spot prices of the DXY’s component currencies. The prices used for the calculation of the DXY are the midpoints between the latest available bid and offer in the component currencies as sourced from ICE Data Derivatives.”

Quarterly changes are calculated as quarterly percentage changes on the index level:

$$r_t^{\text{DXY}} = \frac{P_t}{P_{t-1}} - 1$$

The index is weighted heavily against the euro at 57.6% ([Intercontinental Exchange \(ICE\), 2026](#)) and it should be acknowledged, that the main reason to select the index over Nominal Broad U.S. Dollar Index is the insufficient historical depth availability of the latter (only available from 2006 on FRED).

The panels

All asset class series are consolidated into a consistent geometrically compounded quarterly excess total return format, comparable across different assets, and combined into a single panel dataframe covering 116 quarters from 1997 through 2025 inclusively. The panel is then used as a basis for the entire further analysis.

Similar logic applies to all macro factors, also consolidated into quarterly levels or differences (depending on the individual factor) and combined into a single panel to apply as an explanatory variable matrix for asset Betas calculation.

Research design

Unsmoothing non-traded indices

It is common to assume *appraisal-based indices*, such as NFI-ODCE (*private real estate*) used in the research, to be affected by appraisal anchoring, where the appraisers are behaviourally affected by the previous estimations, hence, the index exhibits autocorrelation and artificially smooth volatility. The observed return value was modelled by D. M. Geltner ([1991](#)) as a weighted average of contemporaneous true return and previous observed return:

$$r_t^* = (1 - \rho) \cdot r_t + \rho \cdot r_{t-1}^*$$

Where ρ is first-order autocorrelation of the returns. The equation allows to restore the true unsmoothed return series with the original volatility (as demonstrated in Figure 2). David Geltner himself empirically estimated ρ as 0.5 for NCREIF Property Index from 1978 through 1990. The present paper uses ρ of 0.8 instead of empirically estimated 0.8609⁷. The unsmoothed index is further compared to Financial Times Stock Exchange Nareit U.S. Real Estate Index Equity REITs index, a public proxy used to validate the correctness of the adjustment. The logic is straightforward: public equities react quicker than private markets, and NFI-ODCE should exhibit correlation with lags of FTSE Nareit. If (after the adjustment) the correlation becomes contemporaneous, the adjustment can be considered viable. Cross-correlation function plots of both the original and unsmoothed returns can be found in Figure 3.

Each of the *private capital* benchmarks exhibits a degree of appraisal smoothing structurally similar to private real estate: despite the general partners' improved understanding of the company cash flows and asset value, there is still an incentive to smooth, since large quarterly changes alarm the limited partners

⁷ Because the latter value results in overcorrection and exploding volatility significantly higher than that of the public proxy (sd 0.23 vs. 0.20).

and affect fundraising. SSPCI partly removes the bias, since the indices are transactions based, but transactions are infrequent and in between transactions the appraisals are still using the general partners valuations, therefore the present observed valuation is affected by the previous, expressed in significant AR(1) autocorrelation (see ρ values immediately below). The present study uses ρ values of 0.3912, 0.5643 and 0.1787, empirically estimated as Pearson correlation for quarterly SSPCI Buyout, Venture Capital and Private Debt percentage total return series from 1997 through 2025 respectively. The lower ρ for private debt reflects the fact the asset class usually has clear contractual cash flows, which are easier to track and evaluate than buyout and venture capital's.

Accounting for *hedge fund* returns is often complicated by the funds' survivorship and self-reporting bias - successful funds tend to self-report, whereas unsuccessful do not, hence a certain exaggeration of returns should be acknowledged. In addition, broad hedge fund indices' returns commonly exhibit autocorrelation due to varying staleness of reported returns: funds applying different strategies hold portfolios with significantly different liquidity, often transacting infrequently and having to rely on best available prices which may be stale quotes from weeks before reporting date. Getmansky et al. (2004) modelled the observed index returns as a moving average of true returns from k previous periods:

$$r_t^* = \theta_0 \cdot r_t + \theta_1 \cdot r_{t-1} + \dots + \theta_k \cdot r_{t-k}, \theta_j \in [0,1], \sum_{j=0}^k \theta_j = 1$$

Barclay Hedge Fund Index exhibits significant first-order autocorrelation. Herein, $k = 2$ is applied to unsmooth the returns with θ_0 of 0.7897, θ_1 of 0.1578 and θ_2 of 0.0525 empirically estimated using raw (monthly) index percentage total returns from 1997 through 2025.

Correlation matrix

As evident from Figure 3, some asset classes may continue exhibiting significant correlation with lags of the public markets even after unsmoothing. In addition, some asset classes may exhibit significant correlation with lags of other indices while having insignificant contemporaneous. These problems prove contemporaneous Pearson correlation insufficient to capture the entire relationship within a quarterly excess returns' granularity. Henceforth, in the present study, yearly excess returns are selected to measure Pearson correlation between asset classes.

APT-style factor framework

The present study's factor model approaches the analysis of asset abnormal returns in style of arbitrage pricing theory (APT) of Stephen K. Ross, drawing from the works of Chen et al. (1986), Ilmanen (2012) and Ang (2014). The choice of factors and the general idea is loosely following BlackRock factor investing framework (BlackRock, 2026), then is further practically applied to the set of asset classes described in the previous sections to enable comprehensive comparison. It should be acknowledged, that different types of alternative instruments are influenced by substantially diverse number of factors, all of which cannot be included in a single model without sacrificing statistical significance on the limited number of observations available (116 quarterly returns from 1997 through 2025). Therefore, the precise explanatory power and specification have to be balanced with the practical limitations to keep the results statistically powerful and comparable across all asset classes, hence the selected set of specific explanatory variables.

The eventual resulting set of factors include *market, commodities, term, credit, liquidity and currency risks*. Several approaches have been tried, including a granular split into numerous related factors such as separate credit and leverage premia (leveraged loans minus treasuries spread serving as credit premium and

high yield bonds minus leveraged loans as leverage/seniority premium) and separate real rate and breakeven inflation (TIPS minus risk-free as real rate and nominal minus TIPS as breakeven), as well as estimating unexpected inflation and orthogonalising some of the factors. However, the practice has shown, that more parsimonious models with fewer, less engineered explanatory variables, produce more stable coefficients and simpler economic interpretation, generally performing better in most regards, taken the sample size (116 observations).

The model itself is an OLS, in which the explained variable is a series of quarterly arithmetic excess total returns of an asset class and the explanatory variables are the K traded macro factors:

$$ER_t = \alpha + \beta_0 F_{1t} + \beta_1 F_{2t} + \cdots + \beta_K F_{Kt} + \varepsilon_t$$

Significance of the estimates are then calculated using Newey-West (1987) estimator with a Bartlett kernel and 4 quarterly lags to produce heteroskedasticity and autocorrelation consistent (HAC) standard errors, overwhelmingly common in empirical finance to account for seasonality and persistent autocorrelation. The number of 4 quarterly lags is selected according to widely referenced Wooldridge (2013) guideline, making both economic and statistical sense, accounting for seasonality and data memory, capturing one full year of persistence in the error term, as well as fitting the proposed formula for $L = 4 \cdot \left(\frac{T}{100}\right)^{\frac{2}{9}}$, which equals 4.13 for 116 quarters of observations.

Two asset classes specifically stand out, namely Bitcoin, and private real estate. The latter shows signs of residual autocorrelation (Figure 2) and cross-correlation with lags of public stocks and public market equivalents (Figure 3) even after unsmoothing, which indicates possible undercorrection and true economic lagged relationship. Coutts et al. (2023) argue that unsmoothing commercial real estate funds returns requires a three-step procedure involving a step with unsmoothing each individual fund, which is impossible in case of aggregated NFI-ODCE data only due to index-level aggregation. It is

acknowledged, that instead the technique is approximated by including Dimson lagged Betas on top of Geltner unsmoothed series. Significance of total Betas is calculated using Wald F-test under Newey-West vcov. Specification for unsmoothed private real estate includes 2 lags of Market, Term and Credit, whereas smooth real estate includes 8 lags. The number of lags was selected with regards to both optimal statistical metrics (see Table 10 and Table 11) and economically viable resulting betas, the latter being a priority, and deduced partly from the ones appearing for the public market equivalent (see Table 12) and from cross-correlation function with the PME, smooth series correlation with PME peaking at the 2nd lag and unsmoothed series keeping significant correlation with 2 lags, justifying 2 lags for unsmoothed series and 8 lags for smooth series implied by ρ of 0.8 (see Table 3). Simple smooth and unsmoothed series with no lags as well as the public proxy ([FTSE Nareit U.S. Equity REITs](#)) are reported for robustness.

Bitcoin, exhibiting strong volatility and a radical change in market regimes, with the beginning large scale adoption during 2020 and only 62 observations to build an OLS, is treated separately and only has contemporaneous Market, Liquidity and Currency factors to test the main hypotheses.

The factor exposure matrix is presented in Table 12. As per arbitrage pricing theory definition, Alpha (or Const. in the table) represent the abnormal return of an asset class, unexplained by exposure to macro factors included. The interpretation stands on the ground of economical sensibilities first and statistical significance second, hence some of the contradictions found in the following paragraphs. Effort is made to reflect and elaborate on each such contradiction. Looking at Commodities movement, which happen to often coincide with reverse movement in Currency, we could argue, that such movements represent commodity- and inflation-related shock, while Term is in theory a total of real rate and breakeven inflation, hence, should be interpreted along these lines.

Findings

Figure 1 represents a simple compounded excess returns trend over the entire sample period (1997-2025). A prominent feature of the plot is the strong consistent outperformance of private capital indices over S&P 500, and a continuous rally of gold.

Tables 2 and 3 feature yearly excess returns and trailing 10-year compounded excess returns accordingly. The tables reveal expectedly strong performance of Bitcoin, as well as buyouts. Venture capital, despite the high returns, is revealed to have underperformed strongly during 2001-2003, 2009 and 2022-2023. Managed futures are shown to exhibit mostly positive returns during crisis years. Commodities, after a strong rally in 2000s, have stopped providing consistently strong returns and started exhibiting frequent reversals.

Table 5 shows a set of statistical properties and tests per each asset class and traded factor. Most of the distributions of excess returns appear to be non-normal, with the exception of gold and treasury bonds, Shapiro-Wilk test failing to reject normality under 0.05 significance threshold - but rejecting at 0.1. Most of the series also appear to be stationary, but gold and Bitcoin should be noted, gold shown to be marginally non-stationary under 0.1 significance threshold by ADF test and Bitcoin rejected to be stationary under 0.05 significance threshold by KPSS test. Bitcoin, commodities, unsmoothed venture capital and unsmoothed real estate all appear to be volatile asset classes, showing annualised volatility greater than 16% on par with large-cap equity (S&P 500) or higher, with large CVaR (24% or more) and Maximum Drawdown (70% or more); Bitcoin and unsmoothed venture capital show strong positive skewness (2.7 or more) together with exceptionally high excess kurtosis (16 or more). Gold, unsmoothed hedge funds, unsmoothed buyout and unsmoothed private debt on the other hand exhibit desirable statistical properties: high annualised average, and no excess CVaR, Maximum Drawdown or annualised volatility over equities or corresponding

public market equivalents, with the exception of unsmoothed Private Debt which does show 10.6% annualised volatility versus 7.9% for Leveraged Loans, but roughly on par with 9.7% for high yield bonds.

Table 6, Figure 6 and Figure 7 show Pearson correlation of yearly excess returns. S&P 500, unsmoothed buyout, unsmoothed hedge funds and NASDAQ Composite immediately fall into a cluster of their own, showing strong (> 0.7) correlation in between each other, with unsmoothed venture capital slightly on the side, strongly correlated to NASDAQ (at 0.73). Unsmoothed private debt, leveraged loans and high yield bonds form a second strongly correlated cluster, closely related to both first and a subset of investment grade bonds, private real estate and listed real estate. Commodities and Bitcoin show low-to-moderate correlation with equities. Gold shows near-zero correlation with most assets. Managed futures and treasury bonds are exceptional, since both exhibit low-to-moderate negative correlation with most other asset classes.

Figure 8 represents trailing 12 quarters rolling correlation of excess returns with S&P 500. Some asset classes, notably real estate and debt-related instruments (private debt, leveraged loans, investment grade bonds) exhibit clear time-dependent patterns, correlation with S&P 500 peaking during 2010s and first half of 2020s. Commodities exhibited a similar trend; however, by 2023 correlation effectively dropped to zero. Since 2023, for the first time in the sample period (1997-2025), treasury bonds became positively correlated with S&P 500; the trend seemingly continues. A notable instrument is managed futures, since its rolling correlation with S&P 500 never crossed ± 0.5 threshold during the sample period.

Tables 12 and 16 represent the estimated Alpha and Beta terms on the selected set of macro factors: market, commodities, term, credit, liquidity and currency. Detailed data and statistical tests on each particular regression can also be found in Table 13, Table 14 and Table 15 in [Appendices](#). According to the present study's proposed model, during the sample period (1997-2025), gold

shows a significant Alpha of 7% annually, positive loadings on commodities and term and negative on currency. Bitcoin demonstrates meaningful loadings on market and currency (after 2020), but the number of observations doesn't allow to draw any strong conclusions. Commodities Optimum Yield futures strategy also has a significant Alpha of 4.6%, an expected positive loading on commodities and negative on term. Managed futures have a 3.5% Alpha, a positive loading on commodities and a marginally significant negative loading on credit. Hedge funds Alpha's marginal significance disappears after unsmoothing, the series showing positive loadings on market, commodities and liquidity. Buyout demonstrates a 5% Alpha, positive loading on market and negative on currency. Venture capital has a strong positive > 1.0 loading on market (consistent with NASDAQ Composite PME); unsmoothing procedure considerably alters the instrument's profile by eliminating significance of positive commodities and currency and negative credit loadings. Private debt has a significant Alpha of 5.5%, positive loadings on commodities, credit and liquidity and a negative loading on currency; the series appears to be considerably different from Leveraged Loans PME. Unsmoothed private real estate with 2 lags of market, term and credit included to calculate total (Dimson) Betas, selected as the most demonstrative private real estate series shows no significant Alpha after unsmoothing, positive loadings on term and credit, and marginally positive loadings on market, commodities and currency.

Discussion

Historical context

The 1997 through 2025 period analysed in the present study covers 29 years of varying market regimes, such the late 90s bull market, the “Lost Decade” of 2000s and the period of growth following the global financial crisis (GFC), as well as several crisis periods (dot-com bubble, the GFC, COVID) and severe geopolitical turmoil (the conflict in Ukraine and the subsequent inflation and gas spikes; the conflict in Gaza).

From contextual perspective, the period provides sufficient diversity and depth for major U.S. institutional investors with 10-year investment horizon prioritised in the present study. From statistical perspective, a sample of 116 quarters (29 full years) of returns per asset class (with the exception of Bitcoin, only covering 62 quarters starting from 2010 Q3), sufficient for statistical significance of betas calculation on a set of explanatory macro factors included for the factor investment framework research.

As evident from Figure 1, Table 2 and Table 3, *private capital* indices have experienced dramatic growth during the last three decades, often outperforming the public market. The high returns are explained by historically low interest rates regime following 2008 global financial crisis, enabling private equity (e.g. *buyout*) funds to use little equity with high leverage to purchase companies, amplifying returns. Expanding multiples also played an important part, allowing the funds to purchase companies at a lower multiple valuation and sell at a higher multiple valuation several years afterwards, or merging smaller companies, effectively implementing multiple arbitrage (“buy and build”) strategy. The following inflow of dry powder enabled rapid scaling of operations, making the private market increasingly more important for major U.S. institutional investors, searching for instruments providing higher returns under the low interest rates.

While buyout showed strong consistent growth, *venture capital* enjoyed similar dynamics and the same cheap financing, albeit, demonstrating higher volatility. The key drivers behind VC growth included an unprecedented rally in the technology sector coupled with the abundance of unicorns, providing returns high enough to cover the losses from investments into the less successful projects, making VC funds a probabilistically profitable business, statistical properties demonstrating very high positive skewness (Table 5). However, VC is a highly cyclical, less reliable vehicle than public stocks and other types of private capital, and in times of spiking volatility, such as the dot-com bubble or COVID, the industry experienced movements of a greater magnitude than stocks, owing to concentrated risk, high leverage and significant illiquidity, denying the ability to rebalance the portfolio timely. Annualised volatility of the index is estimated to exceed 30% as shown in Table 5.

Private debt, on the contrary, proved to be a more stable instrument with remarkably lower volatility, the index properties being comparable to that of high yield bonds with enhanced returns. Following the global financial crisis, stricter banking regulations forced the companies to shift towards private debt. For the investor, the instrument provides a set of benefits, such as floating rate structure providing a consistent premium over the public markets, liquidity premium and middle market exposure. The increase in buyouts fuels the demand for private debt, being the main source of financing for the industry. The senior position of private debt and high upfront fees contribute to higher returns and protection for the investors.

As evident from Figure 1, Table 3 and Table 5, all three private capital indices consistently outperformed their public peers, buyout and venture capital providing higher returns than S&P 500 and private debt outperforming high yield bonds, while each index (with the exception of the highly volatile VC) preserving comparable skewness and volatility. It is worth mentioning though, that the

buyout index exhibits significantly higher excess kurtosis than S&P 500, showing higher risk of tail events.

During the entire period of the present study, *hedge funds* provided somewhat lower returns than equities and private capital, but proved to be consistently less volatile and remarkably stable during different market regimes, exhibiting, however, comparable excess kurtosis (Figure 1, Table 5). Hedge fund returns have been historically driven by high security dispersion, multi-strategy approach, leverage and active management providing management skill Alpha. Hedge funds' structure gives them exposure to numerous asset classes and instruments, enables them to implement unconstrained trading strategy and profit from volatility in the markets, currencies, commodities and interest rates, making them a desirable diversification vehicle for investors seeking protection against uncertainty or simply to outsource active management to the well-performing funds.

Hedge fund indices are considered subject to survivorship bias, which can exceed 2% annually ([Liang, 2000](#)). Barclay Hedge Fund Index, however, is the index least affected by survivorship bias thanks to the volume of the BarclayHedge database including the largest number of both active and defunct funds ([Joenväärä et al., 2021](#)). On the other hand, it should be stated that major U.S. institutional investors usually allocate to funds in the upper performance percentile, both parties benefiting from the dynamics. The above arguably makes survivorship bias concerns less critical in the context of the research.

During the sample period, *managed futures* provided lower returns than the majority of the asset classes. However, the strategy some of the lowest volatility among the asset classes, being comparable to that of treasury bonds (Table 5). Moreover, it emerged as the only asset class giving positive returns during each significant crisis covered (2000, 2008, 2020, 2022), demonstrating performance not always exhibited even by protective assets such as government bonds and gold (Table 2). Commodity trading advisors (CTAs) implement momentum strategies

to systematically profit from following market trends in either direction, capable of providing diversification and crisis Alpha under any market conditions.

Private real estate after Geltner unsmoothing shows a remarkably unique profile, providing considerable return but showing significant volatility comparable to that of public market equity and buyout, high excess kurtosis and skewness specifically pointing to vulnerability to tail events (2008, 2022) (Figure 1, Table 5). During the sample period, real estate thrived on historically low key rates and cap rates, and persistent undersupply of new construction. Real estate typically relies on high amounts of leverage, but provides the investors a stable cash flow, inflation hedge and illiquidity premium, while also serving as a big part of foreign investments in U.S.

During the sample period, *commodities* experienced several very different regimes. From 1990s through 2010 rapid growth and industrialisation of emerging markets created an unprecedented demand for industrial metals as well as all sorts of other resources required to maintain growth. From 2010 through 2020 negative roll yields pushed the demand for scarce assets such as gold, palladium and zinc. 2020 through 2026 were characterised by rapid energy transition, growing geopolitical tension and fragmentation driving high volatility in scarce resources, gold, oil and gas, and incentivised investors to search for hedges against inflation and weakening currency. As evident from Table 5, commodities futures (implementing Deutsche Bank Optimum Yield strategy to minimise negative roll yields) demonstrated high volatility exceeding that of public stocks, while providing lower returns. However, commodities provide significant inflation and geopolitical risk hedges and show rapid rise in prices during times of growth and high demand, hence should be analysed in the context of varying market regimes (Figure 1, Table 2, Table 3).

Gold has long maintained its status as a scarce safe haven asset with limited supply, surging in times of crisis (such as GFC and COVID) and hedging from monetary debasement. During the sample period, gold benefited from low interest

rates reducing opportunity cost of holding (being a non-yielding asset). The latter part of the sample (from 2020 through 2026) exhibited geopolitical instability and massive purchases of gold by countries such as China, Poland and Turkey, pushing the demand. J.P. Morgan (Jury ([2026](#))) remains bullish on gold, pointing at shifting macroeconomic trends. As of 2025 Q4 gold prices experienced a spike, the asset trading at a historically high price, exceeding \$4,000 per ounce (Figure 1). Showing high returns and lower volatility than stocks, excess returns of gold notably maintaining low excess kurtosis and skewness, being the only asset class during the sample period (with the exception of treasury bonds) which is not rejected to be normally distributed under 5% significance level by Shapiro-Wilk test (Table 5).

As of 2026, *digital assets*, represented by *Bitcoin* in the present studies, are still not considered an option for significant allocation by major U.S. institutional investors, the allocations usually staying within 1-5% of the portfolios⁸ (if any). 2024-2026, however, have seen a positive dynamic with regards to institutional adoption with the launch of spot ETFs, increasing acceptance by both the governments and institutions, and the development and major market infrastructure, regulated exchanges and derivative products. Bitcoin in particular has fixed supply, often argued to be a “digital gold”, while also experiencing “halving” events, reducing the reward for Bitcoin mining by halve, stretching the supply even further. Bitcoin historically benefited from eased liquidity conditions, being a volatile asset with considerable media attention, correlated to technology stocks and overall global risk appetite. As evident from Table 5, Bitcoin provided extreme returns and extreme volatility with high positive skewness, arguably being a desirable extreme positive event bet. There has been, however, a gradual decrease in the magnitude of returns and volatility pointing at the growing acceptance and maturity of the industry and, arguably, at the continuation of the

⁸ Citation needed.

aforementioned dynamics towards stabilisation and becoming more equity/commodity-like in statistical properties (Table 2).

Correlations

As evident from Table 6, Figure 6 and Figure 7, all private asset classes exhibit moderate to strong correlation between each other and the public stocks and bonds after unsmoothing. *S&P 500, hedge funds, buyout and NASDAQ* cluster particularly well together, showing strong correlation between each other. Upon examining rolling 12-quarters correlation (Figure 8) and NASDAQ link, *venture capital* adds up to the cluster, all asset classes demonstrating strong correlation throughout the entire sample period.

The second cluster, closely correlated with the previous, consists of *investment grade and high yield bonds, private debt, leveraged loans, listed real estate and unsmoothed private real estate*. Each of the assets is moderately-to-strongly correlated with the other and with the previous cluster (S&P 500, hedge funds, buyout and venture capital), high yield bonds and private debt clustering closer together and with the public markets, while listed and private real estate and investment grade bonds clustering slightly more separately. All five are largely driven by real rate, hence the correlation. Rolling correlation reveals that real estate (both listed and private) correlation to the public market is particularly volatile and tends to increase in times of crisis (2008, 2022 and onwards); investment grade bonds exhibiting similar dynamics.

Commodities and Bitcoin are revealed to be moderately positively correlated with the first clusters (S&P 500, hedge funds and buyout). Rolling correlation shows Bitcoin has been positively correlated with the public markets since the second half of 2010s, while commodities exhibit strong regime dependency and tendency to be affected by crises.

Gold and managed futures are largely uncorrelated with the rest of the market, managed futures showing slightly negative correlation with most other

assets. Gold is significantly negatively correlated with Bitcoin, reflecting the opposite risk profiles of the two, Bitcoin being a speculative volatile asset and gold a safe haven. Rolling correlation reveals apparent mean reversion for both managed futures and gold, but neither exceeds the range of ± 0.5 correlation with the public market within a 12 trailing quarters window.

Finally, the *treasury bonds* show weak-to-moderate negative correlation with most other indices, behaving as expected, being a “risk-free” protective asset class which attracts investors when the rest of the market underperforms.

Factor exposures

Traditional asset classes (Table 12) are mostly reported for robustness, to ensure the model is functioning as intended. According to the model, *investment grade and high yield bonds* returns are mostly a weighted sum of term and credit premia with some currency exchange rate exposure - depreciating DXY coincides with lowering rates and higher global liquidity and makes it cheaper to serve debt for companies which issue debt in USD but earn revenue globally. *Treasury bonds* should be explained by term, albeit the term factor only includes 10Y bonds, hence the difference, the marginally significant (but tiny) betas for market and commodities likely being statistical fluctuation. *S&P 500* appears to fit the broader market nearly perfectly, Pastor-Stambaugh liquidity likely representing the correlation between compositions of S&P 500 and the broader market factor from Fama-French 3 factor model.

Optimum yield commodities futures are largely commodities (Beta ≈ 1) with added exposure to term - negative beta shows reducing inflation, which enables markets backwardation and positive roll yield: the expiring futures can be sold expensively, and the new ones cheaper. The analysis suggests, that the significantly positive Alpha of 4.6% annually is a true managerial alpha attributed to the Deutsche Bank’s optimum yield strategy performance benefits, discussed in [Commodities \(optimum yield commodity futures strategies\)](#), consistent with

claims by C. B. Erb & Harvey (2006) that commodity strategies can bring equity-like returns utilising roll structure, weighting and rebalancing.

Gold, Bitcoin and managed futures are mostly unexplained by the current framework and are considered complicated to model.

Gold, being a protective asset, shows a significant 7% annualised Alpha (contradicting the cautious stance by C. Erb & Harvey (2024)), which is argued to partly represent true safe haven premium and partly an artefact of the sample period (1997-2025), characterised by low interest rates and hence low opportunity costs as well as the increased demand from the central banks. The significant high Betas on commodities (*partly explained by inclusion of gold in BCOM*) and term represent monetary policy, inflation and inflation expectations exposure, while significant negative Beta on currency reflects currency devaluation protection, each perfectly aligning with theory.

Managed futures returns are largely unexplained within the current framework ($R^2 \approx 0.11$), showing a significant Alpha of 3.5% annually. However, it is argued, that the Alpha cannot be claimed to represent reality, since the model omits important factors needed to confirm it. Marginally significant negative Beta on tightening credit spread being a risk-on signal and often accompanying bull markets, while a significant Beta on commodities appreciating reflect successful trend-following positions and inflation hedging. Notably, the insignificant but considerable negative beta on the market could signify the “crisis Alpha” commonly attributed to managed futures, nearly perfectly persistent through 2000-2002, 2008, 2020, 2022 (Table 2).

Bitcoin is not easily comparable to the rest of the asset classes in the series: the price history is only available since 2010, the asset class has transitioned through two distinct market regimes (pre 2020 and post 2020 when mass retail adoption started) and the number of observations (62 quarterly excess total returns) is insufficient to include all factors while preserving meaningful

coefficients not only due to statistical power but also due to spurious correlations. The present study suggests that the two market regimes are particularly distinct since the emerging industry has transitioned into a more mature state, and the two should be analysed separately. Henceforth, only 3 theoretically proposed factors are kept (market, liquidity and currency) and several regressions are presented: naive 2010-2025, dummy, 2010-2019 and 2020-2025 (see Table 16). Naive regression is also reported in Table 12 for demonstration purposes. All Bitcoin regressions appear to be unstable, with few observations and a possibly insufficient set of factors to claim a complete specification, henceforth, since researching Bitcoin is outside the scope of the present study, the following findings and discussion on the digital asset should be considered speculative and included for illustrative purposes.

As shown by going from naive to dummy model, simply introducing a pre 2020 dummy variable immediately eliminates Alpha significance, existence of which is further put into question by separating the pre and post 2020 regressions. While the coefficients for the pre 2020 period are magnitudally meaningless, directionally each makes sense, despite the lack of statistical significance, the market factor has a large Beta, showing the asset could amplify market movements, whereas the currency has a large negative Beta, meaning the asset could be used to escape currency devaluation. The overly large negative liquidity Beta could be interpreted as volatility amplification for the risk-on asset or partly by “flight to novelty” from the worsening liquidity conditions.

The post 2020 period is characterised by dramatically reduced liquidity Beta, seemingly similar betas for market and currency and much lower returns in the magnitudes of hundreds of percent per year rather than thousands. The explanatory power of the regression is also considerably increased, R^2 growing from 0.27 in pre 2020 period to 0.3 in post 2020 with an insignificant annualised Alpha of 26.9%. The study assumes the magnitudes are going to lower even further during the following years due to increasing institutional adoption and

large-scale arbitrage eliminating the inefficiencies which made the previous phase possible. The post 2020 phase returns seem to be largely explained by market movements, Bitcoin moving roughly at 5x the market (see Table 2), which is perfectly captured by the Beta. Some additional currency exposure is marginally significant with the expected sign.

Before moving to private capital, hedge funds and real estate, it should be noted, all series are net-of-fees total return series, meaning the fees are already subtracted and the alphas present actual abnormal return from investor's perspective, albeit could include selection bias. The selection bias primarily concerns buyout and private debt, having very high alphas and being acquired on State Street's client database. Other private asset classes are eventually shown to provide no statistically significant Alpha either after unsmoothing procedure or even without unsmoothing.

Unsmoothed buyouts provide a significant Alpha of 5% annually (taken the inclusion of Fama-French style indicators and separation of credit into credit and leverage premium did not reduce the 5% Alpha as discussed in the previous paragraphs). However, the study does not address selection bias and only partly addresses the illiquidity premium via inclusion of Pastor-Stambaugh traded liquidity factor, making the Alpha calculation inconclusive. Phalippou (2020), however, argues the Alpha is non-existent, proving his point using public market equivalent calculations, which cannot be replicated in the present study due to data limitations. The research agrees, that until Phalippou's approach is precisely replicated and rigorously assessed, the existence of Alpha cannot be proven. The asset class is loading heavily on market and currency factor, the latter likely explained by increasing competitiveness of export-oriented firms, which are frequently targeted by buyout funds. It is also consistent with prior research (Welch & Zhou, 2024):

“We find that a 1% change in the export-weighted exchange rate explains in the OLS full- sample regressions (1) a 0.2% point change in net income scaled by total sales;”

Smooth series is reported for robustness. Unsmoothing reveals the correct patterns, lowering Alpha, amplifying market and currency dependence and removing statistical significance of commodities prices.

Unsmoothed private debt exhibits a statistically significant Alpha of 5.5% annually (which, however, cannot be considered proven for the reasons similar to buyouts’), contradicting the findings by Erel et al. ([2024](#)), claiming positive but insignificant net-of-fees risk-adjusted returns for private debt funds. The unsmoothed series is very similar to the smooth series due to low autocorrelation in the latter ($\rho \approx 0.1787$), resulting, however, in a significant increase in liquidity loading: improving liquidity conditions likely reduces the risk associated with illiquid assets, lowering the required premium, while also reducing cost of redemption and origination of loans given by the private debt funds. Positive loading on commodities and negative on currency is expected: increasing commodities prices and depreciating currency coincide with inflationary pressure and rising rates, moving the floating base rate (associated with most of the issued private debt) and enhancing returns. High positive loading on credit means the assets class is credit-like with the aforementioned premiums on top.

Unsmoothed venture capital is vastly different from the smooth version of the series due to high autocorrelation in the latter ($\rho \approx 0.5643$, see Figure 4 and Figure 5), the unsmoothed series making more economic sense with greatly reduced Alpha (albeit both series’ Alphas are statistically insignificant in the first place), providing a much higher than 1 market beta, meaning venture capital could be described as strongly amplified equities, consistent with Cochrane ([2005](#)).

Unsmoothed hedge funds are different from the smooth series primarily with regards to the unsmoothing procedure lowering the Alpha (significant at 0.05

for smooth series) and erasing its statistical significance (consistent with the conclusion by Dichev & Yu (2011) that the hedge funds Alpha is close to zero), herein it is argued that the broad hedge funds universe does not provide a significant managerial Alpha after accounting for stale pricing and appraisal smoothing. It should be however acknowledged, that major U.S. institutional investors typically have exposure to the higher performing funds, hence it is argued *the actual Alpha in the context of the paper could be higher* than reported in the table and the factor Betas could be different. The asset class has a moderately high beta on the market with additional exposure to commodities and liquidity: the beta on commodities likely reveals the significance of strategies benefitting from momentum, volatility and inflation, while the liquidity beta shows the importance of funding liquidity and arbitrage efficiency.

Private real estate proves to be an asset class, which is likely significantly affected by the majority of factors in the framework and provides no Alpha on its own. Unsmoothing procedure effectively erases Alpha significance even before introduction of lagged variables into the equation. Lagged variables are necessary, since private real estate is revealed to be an inert, slow-reacting asset, its price being greatly affected by market movements up to several quarters before even after unsmoothing (as evident from Figure 3 and Figure 2). The most conclusive series is henceforth R. Est. (unsm.)^c, including 2 lags of market, term and credit factors. A smooth series with 8 lags of the same variables as well as contemporaneous smooth and unsmoothed series and a series of listed real estate serving as a public market equivalent are reported for robustness. High beta on term combined with marginal commodities exposure reflects a loading on stronger economic outlook and strong inflation hedging properties, while high credit beta reflects the increasing rental yields. Finally, marginally significant positive currency depreciation beta could possibly be explained by translation gains for foreign investors in U.S. real estate and by correlation with the overall strength of the economy. The analysis suggests, that private real estate, after

accounting for appraisal smoothing and lagged reaction to economic indicators, is not too different from listed real estate, being in essence an equity-like high bet on term, credit and inflation - a finding perfectly consistent with Pagliari et al. ([2005](#)).

Conclusion

The findings of the present study remarkably correspond with Brown et al. (2024), showing significant buyout Alpha (*more ahead*) and close to zero for real estate on a comparable sample period (1997-2025 in this paper versus 1987-2022 in their study). The hypotheses on insignificance of Alpha for hedge funds, real estate and venture capital after unsmoothing is empirically proven, but could not be proven for buyout and private debt. With a few rare exceptions (such as gold, Bitcoin and managed futures), the majority of the asset classes in the present study can be predicted by market movements and macro factors, largely agreeing with Ang (2014), Ilmanen (2012) and Chen et al. (1986), confirming another hypothesis and proving the naive straightforward approach to diversification largely useless in the long run, unless the allocations are rigorously managed (which in itself is already a “management Alpha”). Managed futures crisis Alpha is observed in the form of positive returns during crises (Table 2) and lack of correlation to public markets (Figure 6), consistent with the evidence by Hurst et al. (2017) that the time series momentum has been positive for 8 out of 10 largest traditional 60/40 portfolio drawdowns from 1880 through 2016. However, the existence of Alpha could not be confirmed statistically (Table 12) since the asset class is largely unexplained by the traded factor framework. Even the asset classes which do seem to provide significant Alphas, such as gold and commodities optimum yield futures (confirming hypotheses on their respective Alphas existence), do not necessarily provide the desirable rates of return, gold lagging behind S&P 500 and commodities futures being highly market regime and cycle dependent (Table 2 and Table 3). Interestingly, private capital indices (buyout, venture capital and private debt) seem to have been significantly outperforming the rest of the market. It should be, however, acknowledged, that the illiquidity premium in buyouts and private debt is only partly addressed via inclusion of Pastor-Stambaugh traded liquidity factor. In addition, the private capital indices originate from a single custodial bank (State Street) reporting the returns of its

own client base, hence, selection bias as well as reporting bias cannot be excluded (researching the biases and verifying using other indices is outside the scope of this paper). Moreover, prior research argues against the existence of the aforementioned Alpha ([Phalippou, 2020](#)).

Some practical allocation recommendations are: recognise the weakening bonds hedge; treat gold and managed futures as a crisis hedge; maintain commodities exposure to have a positive inflation loading; treat buyout and especially venture capital as leveraged equity with illiquidity premium; maintain limited venture capital and Bitcoin positions as a high return bet; recognise real estate to be a bet on term, credit and inflation; allocate using factor risk exposure and plan with regards to the macroeconomic forecast.

Hereby, it is proposed to envision long-term institutional diversification not as a search for outperformance or an optimal momentum portfolio, but also a diversified macro bet, a set of weights allowing the institutional investor a hedge against each specific kind of risk he would expect occurring during the investment horizon.

Limitations and further research

Proposing a framework to realise that vision is the immediate topic for further research. Other such topics include a rigorous validation of private capital indices, which are an especially complicated topic, not fully covered in the present study, hence the influence of related concepts (such as transaction costs, exit opportunities, illiquidity premium, survivorship, selection and reporting biases...) in the context of the present framework is required to be estimated. Market caps of the asset classes and applicability with regards to portfolio size of a major U.S. institutional investor is another such topic. The research notably lacks the analysis of different time periods; splits along the lines of significant market regime changes were also tried preliminary only to show little change for the post 2008 period. A more rigorous analysis, however, remains to be implemented. It should also be acknowledged, that the results of some of the tests on the residuals (namely Ljung-Box, Breusch-Pagan, White and Shapiro) may produce significantly small p-values, pointing at possible violations of complete OLS specification; hence, the resulting magnitudes and functional forms may be inconclusive for some of the assets. Modern modelling techniques and structural models could be used instead of OLS to accomplish period analysis and to provide more robust coefficients estimation. Some of the asset classes (namely treasury bonds, hedge funds, buyout, venture capital and private debt) could use different indices, more common academically and more fit for the task - however, due to limited access to data, the choice was made to stick with the current selection while providing necessary comments.

As stated in the previous sections, different approaches to factor set construction were tried, notably, a non-traded set including complicated economic indicators, such as CPI, Chicago Fed National Activity Index and Chicago Fed Adjusted National Financial Conditions Index. The non-traded macro set produced surprisingly similar results and Alphas, but the framework was arguably

more bloated, included many collinear factors and theoretically did not provide a similarly clear definition for the intercept, therefore a choice was made to switch to a traded factors or “market proxies” framework.

Inside the market proxies, more granular divisions were tried, separating credit factor into credit (leveraged loans minus treasuries) and leverage (HY bonds minus leveraged loans) and term factor into real rate (TIPS minus risk-free) and breakeven inflation (nominal treasuries minus TIPS), while also adding Fama-French style factors (small minus big, high minus low), volatility (VIX delta), approximating unexpected inflation and orthogonalising some of the factors to reduce multicollinearity. In practice, the most parsimonious models with the least number of factors and the fewest complex transformations proved not only to exhibit the best statistical properties and preserve ease of interpretation, but also produced largely similar coefficients (including the intercepts), so the judgement was made in favour of their sufficiency at the current scope and time granularity.

Additionally, a custom method calculating implied correlation as a signed square roots of R^2 from a distributed lag model predicting an asset class’s excess total returns from another asset class’s excess total returns was tried as an alternative to traditional contemporaneous Pearson correlation:

$$\begin{aligned}\rho_{(r^{\text{real estate}}, r^{\text{S\&P 500}})}^{\text{implied}} &= \text{sgn}\left(\sum_{k=0}^K \beta_k\right) \sqrt{R^2}, R^2 \leftarrow r_t^{\text{real estate}} \\ &= \alpha + \beta_0 \cdot r_t^{\text{S\&P 500}} + \beta_1 \cdot r_{t-1}^{\text{S\&P 500}} + \dots + \beta_K \cdot r_{t-K}^{\text{S\&P 500}}\end{aligned}$$

However, the method required additional workarounds for cases with differently signed lags, while a yearly returns Pearson correlation proved to be sufficient.

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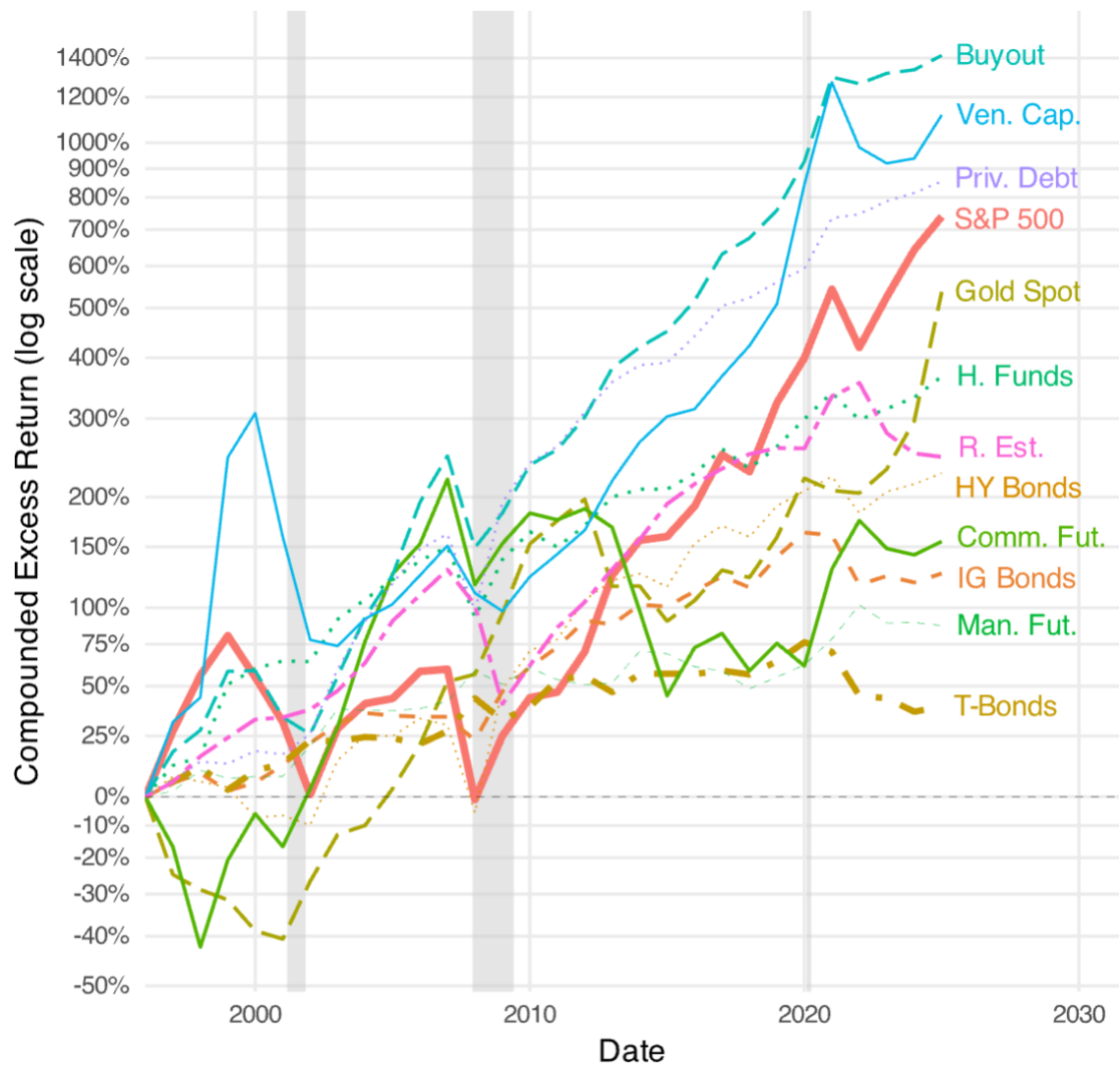
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Appendices

Table 1: Asset class indices

Asset Class	Index	Type
<i>Traditional (equity and fixed income):</i>		
Public equity	S&P 500®	Total Return
Investment grade bonds	ICE BofA US Corporate Index	Total Return
High yield bonds	ICE BofA US High Yield Index	Total Return
Treasury Bonds	Swinkels (2019) approximated U.S. aggregated treasury bonds total return with static weights per maturity in the style of Bloomberg US Treasury Total Return Unhedged USD (LUATTRUU)	Total Return
1-Month Treasury Bill (risk-free rate)	Ibbotson Associates (through May 2024) / ICE BofA (from June 2024) US 1-Month Treasury Bill Index from Kenneth R. French Data Library	Simple Periodic Return
<i>Commodities, precious metals, digital assets:</i>		
Gold (spot price)	London Bullion Market Association Gold Price PM Fix	Spot Price
Digital assets (Bitcoin)	Coin Metrics Bletchley Index Bitcoin - Total Returns	Total Return
Diversified commodities	DBIQ Optimum Yield Diversified Commodity Index	Total Return
<i>Hedge funds:</i>		
Hedge funds	Barclay Hedge Fund Index	Net-of-Fees Total Return
Managed futures	Barclay BTOP50	Net-of-Fees Total Return
<i>Private capital:</i>		
Buyout	State Street Private Capital Indices: Buyout	Net-of-Fees Total Return
Venture capital	State Street Private Capital Indices: Venture Capital	Net-of-Fees Total Return
Venture capital PME	NASDAQ Composite Index	Total Return
Private debt	State Street Private Capital Indices: Private Debt	Net-of-Fees Total Return
Private debt PME	Morningstar LSTA US Leveraged Loan Index	Total Return
<i>Real estate:</i>		
Private real estate	National Council of Real Estate Investment Fiduciaries Fund Index - Open End Diversified Core Equity	Net-of-Fees Total Return
Listed real estate	Financial Times Stock Exchange Nareit U.S. Real Estate Index Equity REITs	Total Return



Line width represents 2025 Q4 log market cap. Bitcoin omitted from the chart.

Figure 1: Compounded yearly excess returns

Table 2: Yearly excess returns

Year	Traditional				Commodities			Hedge funds		Private capital					Real estate	
	S&P 500	IG Bonds	HY Bonds	T-Bonds	Gold Spot	Bitcoin	Comm. Fut.	Man. Fut.	H. Funds	Buyout	Ven. Cap.	Nasdaq Comp.	Priv. Debt	Lev. Loans	R. Est.	Listed R. Est.
1997	27%	5%	8%	6%	-25%	-	-17%	2%	12%	18%	31%	16%	6%	1%	6%	13%
1998	23%	4%	-2%	6%	-5%	-	-31%	8%	3%	8%	9%	33%	7%	1%	10%	-21%
1999	16%	-6%	-2%	-8%	-4%	-	38%	-3%	30%	24%	141%	77%	-1%	-0%	7%	-9%
2000	-14%	3%	-10%	7%	-11%	-	18%	1%	6%	0%	18%	-43%	5%	-1%	7%	19%
2001	-15%	7%	1%	3%	-3%	-	-11%	0%	3%	-16%	-36%	-24%	-1%	0%	1%	10%
2002	-23%	8%	-3%	9%	24%	-	24%	12%	-0%	-6%	-31%	-33%	8%	0%	3%	2%
2003	27%	7%	27%	-0%	19%	-	26%	14%	17%	24%	-2%	49%	27%	9%	7%	36%
2004	10%	4%	10%	2%	3%	-	37%	-0%	8%	24%	11%	8%	19%	4%	11%	30%
2005	2%	-1%	-0%	-0%	14%	-	28%	-0%	7%	17%	5%	-1%	15%	2%	17%	9%
2006	10%	-0%	7%	-2%	18%	-	12%	1%	7%	31%	11%	5%	13%	2%	10%	29%
2007	1%	-0%	-2%	5%	26%	-	27%	3%	5%	18%	11%	6%	6%	-2%	10%	-19%
2008	-38%	-8%	-28%	13%	3%	-	-32%	12%	-23%	-29%	-16%	-41%	-25%	-30%	-12%	-39%
2009	26%	20%	57%	-8%	25%	-	16%	-5%	24%	14%	-6%	45%	49%	51%	-30%	28%
2010	15%	9%	15%	5%	29%	250%	12%	6%	11%	19%	13%	18%	16%	10%	15%	28%
2011	2%	7%	4%	10%	9%	1,321%	-2%	-4%	-6%	6%	9%	-1%	6%	1%	15%	8%
2012	16%	10%	16%	1%	8%	318%	4%	-2%	8%	13%	9%	17%	13%	10%	10%	18%
2013	32%	-1%	7%	-6%	-27%	7,183%	-7%	1%	11%	20%	20%	40%	12%	5%	13%	2%
2014	14%	8%	3%	7%	0%	-59%	-26%	12%	3%	8%	15%	15%	6%	2%	11%	30%
2015	1%	-1%	-5%	0%	-12%	-4%	-27%	-1%	0%	6%	10%	7%	1%	-1%	14%	3%
2016	12%	6%	17%	-0%	8%	107%	19%	-5%	6%	12%	3%	9%	10%	10%	8%	8%
2017	21%	6%	7%	1%	12%	1,319%	5%	-2%	9%	19%	13%	29%	12%	3%	6%	4%
2018	-6%	-4%	-4%	-1%	-3%	-62%	-13%	-6%	-7%	6%	12%	-5%	3%	-1%	5%	-6%
2019	29%	12%	12%	5%	16%	74%	11%	4%	8%	11%	17%	34%	6%	6%	2%	23%
2020	18%	9%	6%	7%	24%	158%	-8%	5%	11%	20%	55%	44%	5%	3%	-0%	-8%
2021	29%	-1%	5%	-3%	-4%	198%	43%	10%	10%	36%	46%	22%	20%	5%	21%	43%
2022	-19%	-17%	-12%	-15%	-1%	-71%	20%	13%	-9%	-2%	-21%	-33%	2%	-2%	5%	-25%
2023	20%	3%	8%	-1%	9%	118%	-10%	-6%	4%	4%	-6%	38%	5%	8%	-17%	8%
2024	19%	-2%	3%	-5%	19%	138%	-2%	0%	4%	1%	2%	23%	3%	4%	-7%	3%
2025	13%	3%	4%	2%	61%	-16%	5%	-1%	8%	5%	17%	16%	4%	2%	-1%	-1%

Table 3: Trailing 10-year compounded excess returns

Year	Traditional				Commodities			Hedge funds		Private capital					Real estate	
	S&P 500	IG Bonds	HY Bonds	T-Bonds	Gold Spot	Bitcoin	Comm. Fut.	Man. Fut.	H. Funds	Buyout	Ven. Cap.	Nasdaq Comp.	Priv. Debt	Lev. Loans	R. Est.	Listed R. Est.
2006	58%	34%	33%	21%	21%	-	153%	38%	138%	194%	125%	35%	148%	19%	110%	168%
2007	25%	27%	21%	20%	103%	-	285%	39%	123%	196%	91%	23%	147%	15%	118%	91%
2008	-37%	13%	-11%	28%	120%	-	277%	44%	67%	95%	47%	-46%	74%	-21%	74%	49%
2009	-31%	44%	43%	29%	186%	-	219%	41%	58%	79%	-43%	-55%	161%	21%	13%	109%
2010	-7%	53%	84%	26%	313%	-	201%	49%	65%	113%	-45%	-8%	187%	34%	22%	124%
2011	12%	54%	91%	36%	363%	-	231%	43%	52%	167%	-6%	19%	209%	36%	39%	121%
2012	69%	57%	128%	27%	306%	-	179%	25%	65%	220%	49%	108%	225%	48%	48%	155%
2013	75%	44%	93%	20%	148%	-	108%	10%	57%	210%	83%	96%	187%	44%	56%	93%
2014	82%	49%	81%	26%	141%	-	12%	24%	50%	169%	91%	108%	156%	40%	57%	93%
2015	81%	49%	73%	27%	85%	-	-36%	23%	39%	144%	99%	125%	125%	36%	54%	83%
2016	83%	58%	90%	29%	70%	-	-32%	17%	38%	109%	84%	132%	118%	47%	50%	54%
2017	120%	67%	108%	25%	50%	-	-43%	12%	43%	110%	86%	183%	131%	56%	45%	99%
2018	233%	75%	175%	9%	43%	-	-27%	-6%	73%	211%	148%	357%	215%	121%	74%	204%
2019	239%	64%	96%	24%	32%	11,610,497%	-31%	3%	51%	204%	208%	321%	124%	55%	155%	194%
2020	247%	64%	80%	27%	27%	8,545,932%	-43%	1%	51%	204%	321%	415%	104%	45%	122%	110%
2021	338%	51%	82%	11%	12%	1,793,740%	-17%	16%	76%	292%	464%	535%	131%	50%	133%	178%
2022	205%	14%	38%	-7%	2%	125,815%	-4%	34%	47%	240%	307%	260%	107%	34%	123%	76%
2023	177%	19%	38%	-2%	54%	3,677%	-7%	24%	38%	195%	220%	254%	94%	37%	65%	86%
2024	190%	9%	39%	-13%	83%	21,625%	23%	11%	40%	177%	183%	280%	88%	40%	37%	48%
2025	223%	13%	52%	-11%	234%	18,997%	76%	11%	51%	175%	202%	313%	94%	43%	19%	41%

Table 4: Unsmoothing methods

Asset Class	Frequency	Method	Specification	Parameters
Hedge Funds	Monthly	Getmansky, Lo, Makarov (2004)	MA(2)	θ_0 : 0.7897 θ_1 : 0.1578 θ_2 : 0.0525
Buyout	Quarterly	Geltner (1993)	AR(1)	ρ : 0.3913
Venture Capital	Quarterly	Geltner (1993)	AR(1)	ρ : 0.5643
Private Debt	Quarterly	Geltner (1993)	AR(1)	ρ : 0.1787
Real Estate	Quarterly	Geltner (1993)	AR(1)	ρ : 0.8000

All parameters acquired empirically for 1997-2025 quarterly returns. Real estate ρ of 0.8 is selected to match PME.

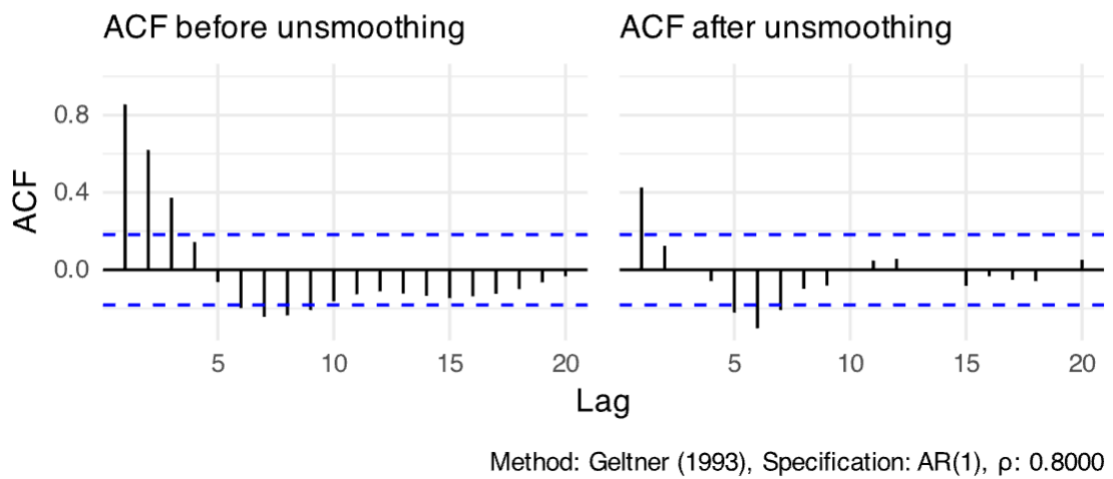
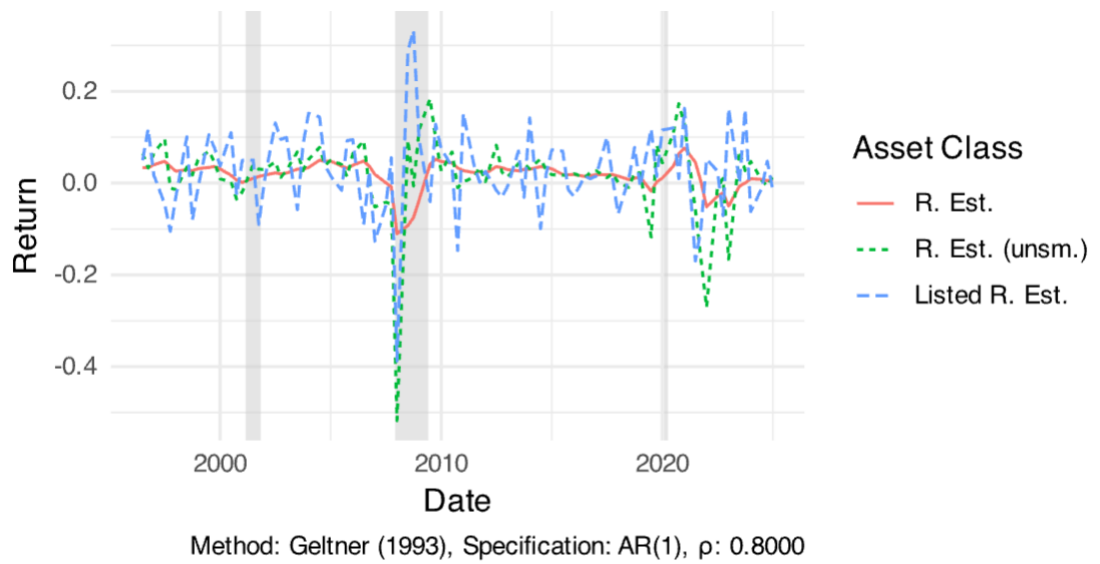
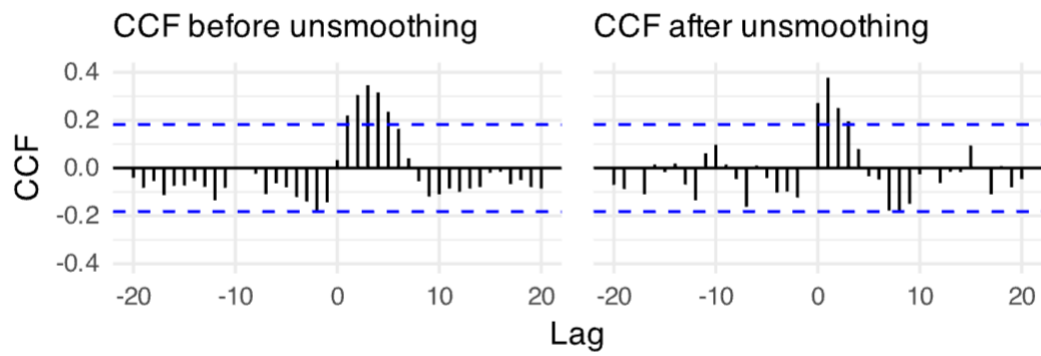
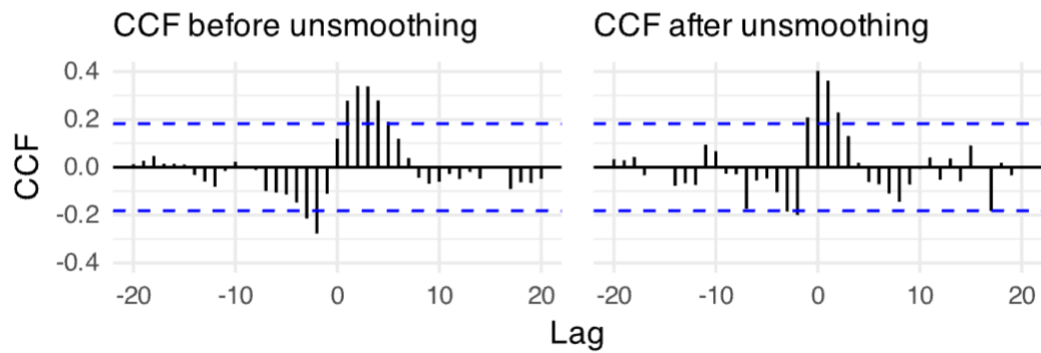


Figure 2: Real estate quarterly returns before and after unsmoothing

CCF With S&P 500



With public market equivalent (FTSE Nareit U.S. Equity REITs)



Method: Geltner (1993), Specification: AR(1), ρ : 0.8000

Figure 3: Correlation of real estate quarterly returns with lags of public stocks

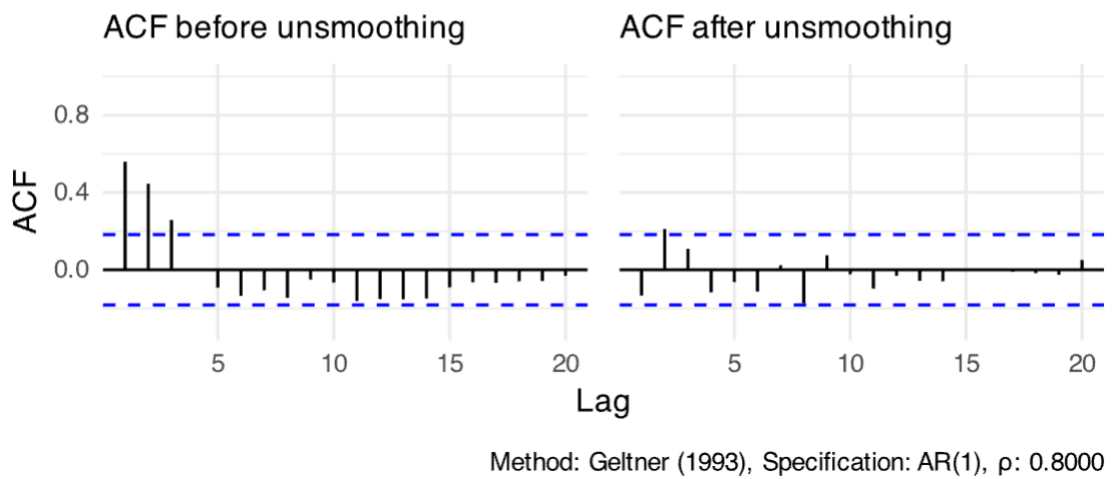
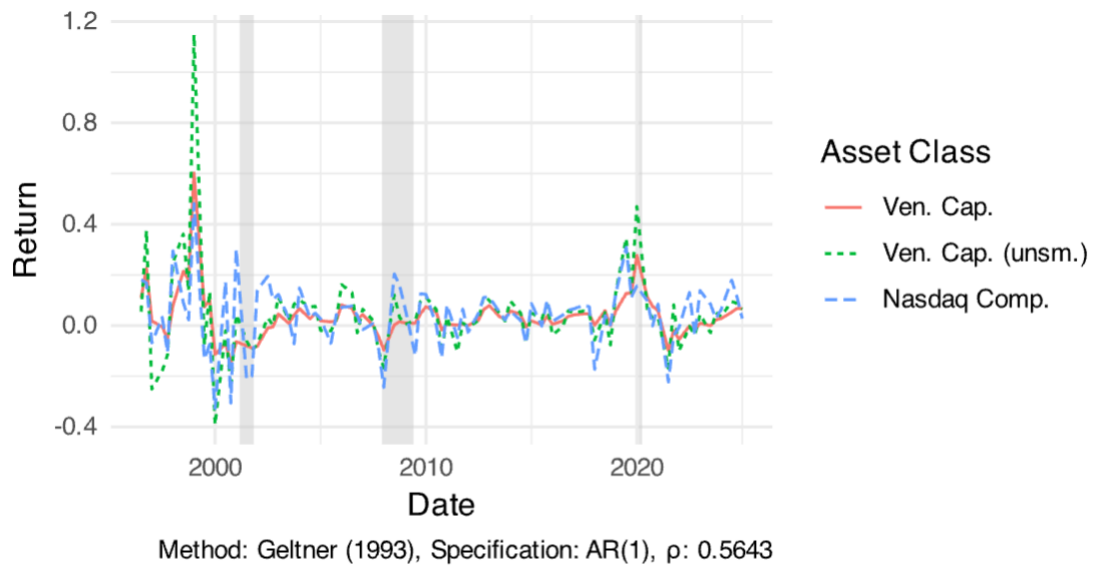
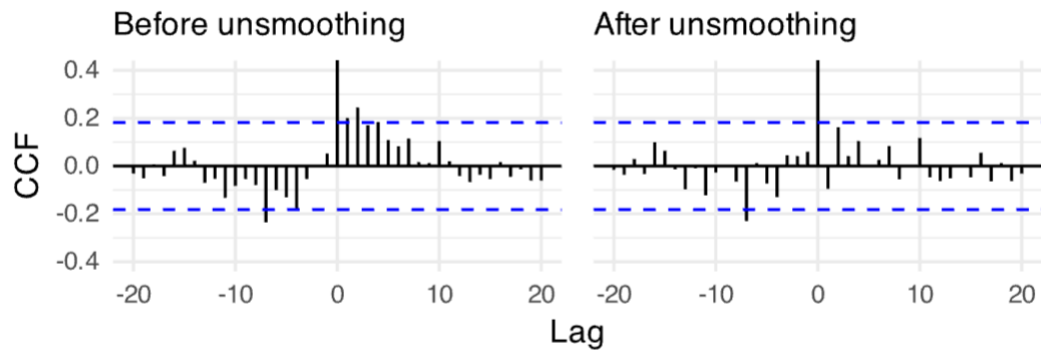
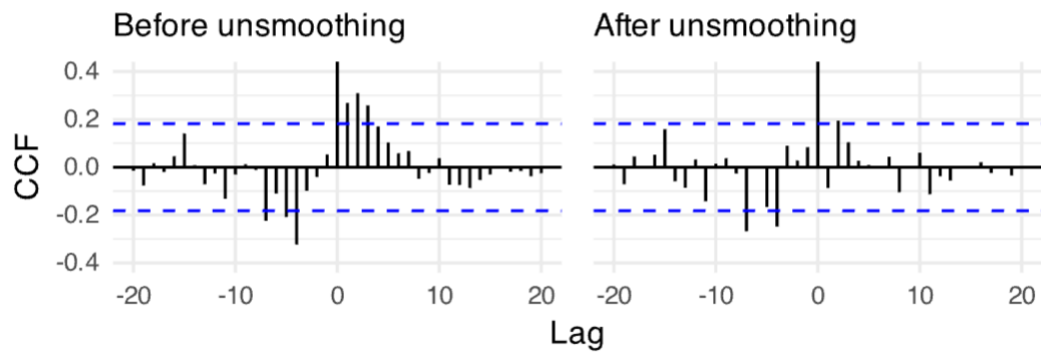


Figure 4: Venture capital returns before and after unsmoothing

CCF with S&P 500



CCF with public market equivalent (NASDAQ Composite)



Method: Geltner (1993), Specification: AR(1), ρ : 0.8000

Figure 5: Correlation of venture capital quarterly returns with lags of public stocks

Table 5: Quarterly excess returns statistics

Asset Class	Metrics			Distribution							Normality	Stationarity	
	Sharpe Ratio	CVaR (0.95 conf.)	Maximum Drawdown	Arithm. Mean	Stdev.	Arithm. Mean Ann.	Stdev. Ann.	Excess Kurtosis	Skewness	Density Plot	Shapiro-Wilk P-val.	ADF P-val.	KPSS P-val.
<i>Traditional:</i>													
S&P 500	0.2639	-0.1753	0.5168	0.0220	0.0835	0.0909	0.1670	0.3701	-0.5991		0.0068	0.0100	0.1000
IG Bonds	0.2531	-0.0614	0.1989	0.0075	0.0297	0.0303	0.0594	1.6492	-0.0311		0.0062	0.0100	0.1000
HY Bonds	0.2363	-0.1080	0.3026	0.0114	0.0484	0.0464	0.0968	5.1979	0.0451		0.0000	0.0100	0.1000
T-Bonds	0.1119	-0.0512	0.2351	0.0033	0.0293	0.0133	0.0586	0.5795	0.4148		0.0757	0.0100	0.1000
<i>Commodities:</i>													
Gold Spot	0.2608	-0.1343	0.4329	0.0186	0.0715	0.0765	0.1430	0.6970	-0.3999		0.0763	0.0611	0.1000
Bitcoin	0.3779	-0.5074	0.7384	0.5802	1.5352	5.2352	3.0704	15.7878	3.6910		0.0000	0.0100	0.0325
Comm. Fut.	0.1332	-0.2417	0.7275	0.0133	0.1001	0.0543	0.2002	1.5569	-0.6525		0.0087	0.0100	0.1000
<i>Hedge funds:</i>													
Man. Fut.	0.1530	-0.0626	0.1876	0.0062	0.0407	0.0250	0.0814	1.8284	0.8432		0.0011	0.0100	0.1000
H. Funds	0.3347	-0.0946	0.2286	0.0143	0.0426	0.0584	0.0852	1.5301	-0.3990		0.0028	0.0100	0.1000
H. Funds (unsm.)	0.3013	-0.1017	0.2319	0.0141	0.0466	0.0576	0.0932	1.8241	-0.3005		0.0052	0.0100	0.1000
<i>Private capital:</i>													
Buyout	0.4971	-0.0996	0.3228	0.0250	0.0502	0.1038	0.1004	1.9858	-0.7208		0.0026	0.0100	0.1000
Buyout (unsm.)	0.3224	-0.1671	0.3896	0.0244	0.0757	0.1012	0.1514	1.5969	-0.5259		0.0045	0.0100	0.1000
Ven. Cap.	0.2916	-0.1167	0.6442	0.0250	0.0856	0.1038	0.1712	14.7578	2.5814		0.0000	0.0100	0.1000
Ven. Cap. (unsm.)	0.1483	-0.2720	0.8251	0.0241	0.1622	0.0999	0.3244	17.5326	2.6858		0.0000	0.0100	0.1000
Nasdaq Comp.	0.2363	-0.2663	0.7667	0.0293	0.1241	0.1225	0.2482	1.4485	-0.2056		0.0069	0.0100	0.1000
Priv. Debt	0.4643	-0.0972	0.2606	0.0206	0.0444	0.0850	0.0888	3.6237	-0.2998		0.0000	0.0100	0.1000
Priv. Debt (unsm.)	0.3885	-0.1187	0.2995	0.0206	0.0530	0.0850	0.1060	3.7836	-0.1345		0.0000	0.0100	0.1000
Lev. Loans	0.2011	-0.0975	0.3257	0.0079	0.0395	0.0320	0.0790	16.4555	-1.0711		0.0000	0.0100	0.1000

Asset Class	Metrics			Distribution							Normality	Stationarity	
	Sharpe Ratio	CVaR (0.95 conf.)	Maximum Drawdown	Arithm. Mean	Stdev.	Arithm. Mean Ann.	Stdev. Ann.	Excess Kurtosis	Skewness	Density Plot	Shapiro-Wilk P-val.	ADF P-val.	KPSS P-val.
<i>Real estate:</i>													
R. Est.	0.3555	-0.0902	0.3926	0.0113	0.0318	0.0460	0.0636	6.0149	-2.0002		0.0000	0.0100	0.1000
R. Est. (unsm.)	0.1323	-0.2522	0.7074	0.0107	0.0808	0.0435	0.1616	16.7145	-3.0611		0.0000	0.0100	0.1000
Listed R. Est.	0.1987	-0.2393	0.6709	0.0199	0.1000	0.0820	0.2000	3.2918	-0.7414		0.0000	0.0100	0.1000
<i>Factors (traded):</i>													
Market	0.2583	-0.1821	0.4997	0.0227	0.0877	0.0939	0.1754	0.3163	-0.5303		0.0125	0.0100	0.1000
Commodities	0.0333	-0.2037	0.7346	0.0028	0.0852	0.0112	0.1704	1.7740	-0.5159		0.0062	0.0100	0.1000
Term	0.1081	-0.0703	0.3038	0.0045	0.0412	0.0181	0.0824	0.5616	0.4486		0.0358	0.0100	0.1000
Credit	0.1665	-0.1376	0.3928	0.0094	0.0567	0.0381	0.1134	5.9675	-0.4436		0.0000	0.0100	0.1000
Liquidity	0.1517	-0.1772	0.5932	0.0134	0.0883	0.0547	0.1766	1.3144	0.2068		0.0586	0.0100	0.0100
Currency	0.0431	-0.0793	0.3988	0.0018	0.0408	0.0072	0.0816	-0.3467	-0.0068		0.8422	0.0100	0.1000

P-values rounded to 4 digits after decimal point.

Table 6: Yearly excess returns correlation matrix

Asset Class	S&P 500	IG Bonds	HY Bonds	T-Bonds	Gold Spot	Bitcoin	Comm. Fut.	Man. Fut.	H. Funds (unsm.)	Buyout (unsm.)	Ven. Cap. (unsm.)	Nasdaq Comp.	Priv. Debt (unsm.)	Lev. Loans	R. Est. (unsm.)	Listed R. Est.
S&P 500		.43	.66	-.27	.02	.39	.11	-.20	.72	.78	.40	.87	.58	.57	.51	.51
IG Bonds	.43		.73	.37	.32	-.09	-.01	-.18	.43	.34	-.08	.31	.59	.65	.29	.56
HY Bonds	.66	.73		-.28	.28	.12	.27	-.27	.67	.69	.11	.57	.91	.93	.29	.62
T-Bonds	-.27	.37	-.28		.11	-.17	-.38	.19	-.36	-.42	-.28	-.32	-.37	-.41	-.02	-.07
Gold Spot	.02	.32	.28	.11		-.48	.27	-.01	.13	.13	.02	.09	.22	.23	-.01	.05
Bitcoin	.39	-.09	.12	-.17	-.48		-.10	-.10	.24	.32	.20	.33	.27	.10	.12	-.08
Comm. Fut.	.11	-.01	.27	-.38	.27	-.10		.00	.47	.52	.23	.11	.45	.30	.35	.37
Man. Fut.	-.20	-.18	-.27	.19	-.01	-.10	.00		-.25	-.17	-.20	-.22	-.14	-.34	-.05	-.05
H. Funds (unsm.)	.72	.43	.67	-.36	.13	.24	.47	-.25		.80	.64	.78	.64	.64	.47	.48
Buyout (unsm.)	.78	.34	.69	-.42	.13	.32	.52	-.17	.80		.48	.73	.76	.63	.58	.60
Ven. Cap. (unsm.)	.40	-.08	.11	-.28	.02	.20	.23	-.20	.64	.48		.73	.01	.07	.19	-.06
Nasdaq Comp.	.87	.31	.57	-.32	.09	.33	.11	-.22	.78	.73	.73		.44	.48	.33	.25
Priv. Debt (unsm.)	.58	.59	.91	-.37	.22	.27	.45	-.14	.64	.76	.01	.44		.91	.41	.67
Lev. Loans	.57	.65	.93	-.41	.23	.10	.30	-.34	.64	.63	.07	.48	.91		.28	.55
R. Est. (unsm.)	.51	.29	.29	-.02	-.01	.12	.35	-.05	.47	.58	.19	.33	.41	.28		.64
Listed R. Est.	.51	.56	.62	-.07	.05	-.08	.37	-.05	.48	.60	-.06	.25	.67	.55	.64	

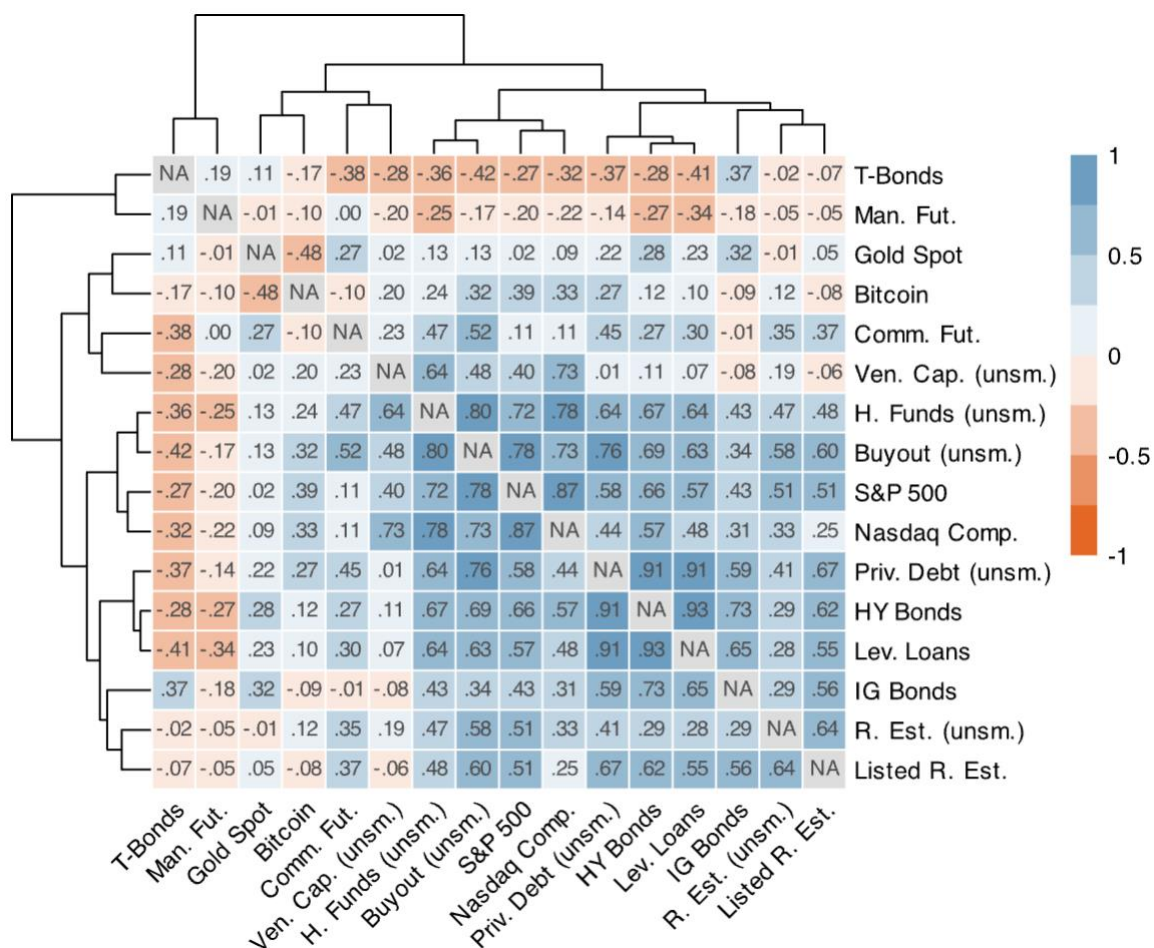
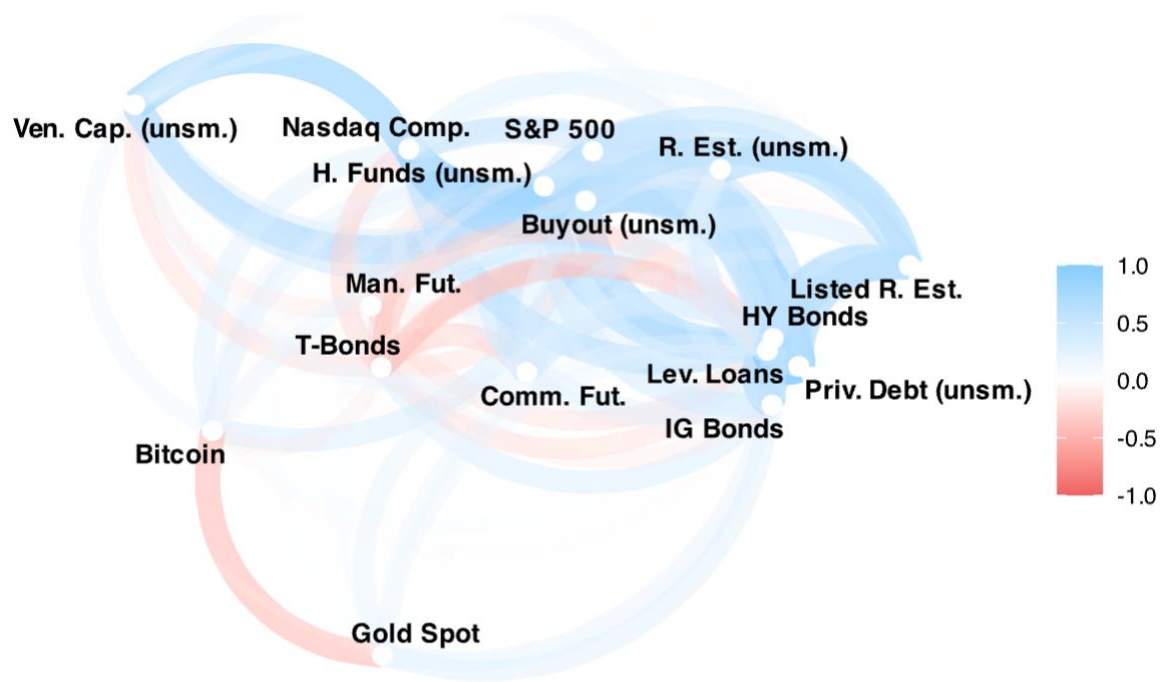
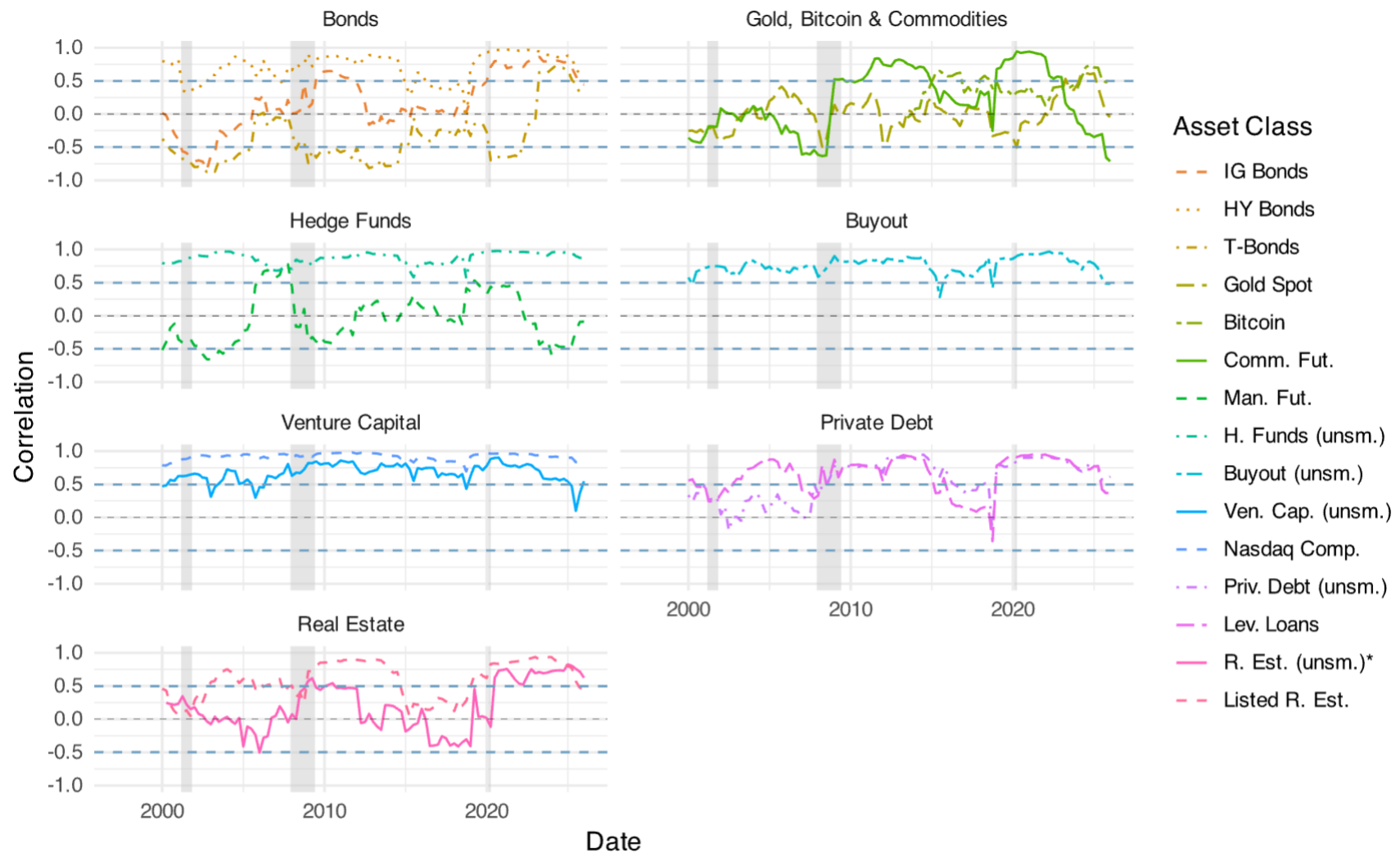


Figure 6: Yearly excess returns correlation matrix



X and Y axis represent principal coordinate analysis dimensions.

Figure 7: Yearly excess returns correlation network plot



Blue dashed lines represent 10% two-tail significance threshold.
 *Real estate correlation is measured against the first lag of S&P 500; all other series against contemporaneous.

Figure 8: Rolling 12-quarters (trailing) correlation of quarterly excess returns with S&P 500

Table 7: Macro factors description

Factor	Index	Interpretation
Market	Fama-French Excess Return on the Market	Market expanding
Commodities	BCOM price return	Commodities prices rising
Term	10Y T-Bonds TR minus 1M T-Bill TR	Term spread widening
Credit	HY Bonds TR minus 3-5Y T-Bonds TR	Credit spread widening
Liquidity	Pastor-Stambaugh traded liquidity	Traded liquidity increasing
Currency	ICE U.S. Dollar Index level percentage change	Dollar strengthening

Table 8: Macro factors quarterly correlation matrix

Factor	Market	Commodities	Term	Credit	Liquidity	Currency
Market		.27	-.41	.76	.25	-.21
Commodities	.27		-.34	.47	.25	-.30
Term	-.41	-.34		-.58	-.19	-.15
Credit	.76	.47	-.58		.29	-.16
Liquidity	.25	.25	-.19	.29		-.04
Currency	-.21	-.30	-.15	-.16	-.04	

Table 9: Macro factors quarterly VIF

Factor	Tolerance	VIF
Market	0.39	2.57
Commodities	0.66	1.52
Term	0.58	1.72
Credit	0.29	3.49
Liquidity	0.89	1.12
Currency	0.78	1.29

Table 10: Optimal number of lags for real estate before unsmoothing

N. lags	R ²	Adj. R ²	Log likelihood	AIC	BIC
0	0.15	0.10	245.10	-474.20	-452.17
1	0.18	0.11	244.84	-467.67	-437.48
2	0.26	0.17	248.24	-468.48	-430.18
3	0.36	0.26	254.02	-474.04	-427.67
4	0.47	0.37	262.05	-484.10	-429.73
5	0.55	0.44	267.58	-489.16	-426.84
6	0.61	0.50	272.78	-493.56	-423.35
7	0.65	0.53	276.33	-494.67	-416.62
8	0.67	0.55	276.85	-489.70	-403.87
9	0.68	0.53	274.41	-478.82	-385.27
10	0.69	0.52	272.81	-469.62	-368.41
11	0.71	0.54	274.06	-466.12	-357.31
12	0.74	0.56	276.11	-464.21	-347.86

Table 11: Optimal number of lags for real estate after unsmoothing

N. lags	R ²	Adj. R ²	Log likelihood	AIC	BIC
0	0.16	0.12	138.13	-260.26	-238.23
1	0.35	0.29	150.63	-279.26	-249.07
2	0.45	0.39	158.87	-289.74	-251.44
3	0.52	0.45	164.85	-295.69	-249.33
4	0.56	0.48	168.16	-296.32	-241.95
5	0.58	0.48	167.93	-289.86	-227.54
6	0.59	0.47	167.87	-283.75	-213.54
7	0.63	0.50	171.14	-284.28	-206.23
8	0.63	0.49	170.06	-276.12	-190.30
9	0.64	0.48	169.57	-269.15	-175.60
10	0.66	0.48	170.02	-264.03	-162.82
11	0.70	0.53	175.11	-268.21	-159.40
12	0.73	0.55	178.75	-269.50	-153.14

Table 12: Quarterly excess returns factor exposure matrix (1997-2025)

Asset Class	R ²	Const.		Betas					
		Const.	Const. Ann.	Market	Commodities	Term	Credit	Liquidity	Currency
<i>Traditional:</i>									
S&P 500	0.99	0.0008	0.0032	0.9419 ***	-0.0009	0.0136	0.0239	-0.0276 **	-0.0158
IG Bonds	0.84	-0.0000	-0.0002	0.0111	-0.0016	0.7132 ***	0.4456 ***	0.0081	-0.0977 *
HY Bonds	0.98	0.0004	0.0017	-0.0066	-0.0060	0.4360 ***	0.9879 ***	0.0004	-0.0586 **
T-Bonds	0.99	0.0001	0.0004	0.0074 .	-0.0083 .	0.7012 ***	-0.0077	-0.0011	-0.0055
<i>Commodities:</i>									
Gold Spot	0.37	0.0176 *	0.0706 *	-0.0256	0.3077 ***	0.6018 **	0.0493	-0.1114	-0.5384 ***
Bitcoin ^a	0.09	0.3991 .	1.5966 .	4.9769 *	-	-	-	-4.8164	-2.8209
Comm. Fut.	0.85	0.0116 **	0.0463 **	-0.0178	1.0219 ***	-0.2825 **	-0.0419	0.0709 .	-0.0235
<i>Hedge funds:</i>									
Man. Fut.	0.11	0.0080 *	0.0322 *	-0.0152	0.1207 *	0.0909	-0.1829 .	-0.0348	-0.0165
H. Funds	0.81	0.0045 *	0.0182 *	0.3579 ***	0.0970 **	0.0207	0.0612	0.0494 *	0.0039
H. Funds (unsm.)	0.82	0.0029	0.0118	0.4129 ***	0.0900 **	0.0548	0.0445	0.0654 **	-0.0195
<i>Private capital:</i>									
Buyout	0.60	0.0174 ***	0.0697 ***	0.3557 ***	0.1448 **	-0.0713	-0.0285	-0.0046	-0.1584 .
Buyout (unsm.)	0.63	0.0119 **	0.0475 **	0.4916 ***	0.0788	-0.0987	0.1578	0.0476	-0.2945 *
Ven. Cap.	0.40	0.0117	0.0469	0.8649 ***	0.2557 **	-0.0892	-0.8113 *	0.0221	0.3788 *
Ven. Cap. (unsm.)	0.39	-0.0014	-0.0056	1.3397 ***	-0.0141	-0.3710	-0.7012	0.2290	0.1851
Nasdaq Comp.	0.85	-0.0002	-0.0007	1.3682 ***	-0.1559 *	-0.1691	-0.1616	0.0823	0.0698
Priv. Debt	0.57	0.0151 ***	0.0602 ***	0.0046	0.0996 **	0.1433	0.4782 ***	0.0291	-0.2095 *
Priv. Debt (unsm.)	0.55	0.0138 ***	0.0552 ***	-0.0015	0.1031 *	0.1696	0.5612 ***	0.0691 *	-0.2452 *
Lev. Loans	0.80	0.0025	0.0098	-0.0817 .	0.0357	0.0789 *	0.6981 ***	0.0272	-0.0373
<i>Real estate:</i>									
R. Est.	0.15	0.0108 .	0.0431 .	0.1327	0.1353 *	-0.0478	-0.3254 .	0.0177	0.1167 .
R. Est. (unsm.)	0.16	0.0045	0.0179	0.1585	0.2989 *	0.1131	0.1393	-0.0472	0.3376
Listed R. Est.	0.53	-0.0012	-0.0047	0.4981 **	0.2107 *	0.8902 ***	0.7542 ***	-0.1522	0.0670
R. Est. ^b	0.67	-0.0087	-0.0347	0.4288 **	0.0730 *	0.8756 ***	0.5711 ***	0.0622 *	-0.0589
R. Est. (unsm.) ^c	0.45	-0.0119	-0.0475	0.4484 .	0.1751 .	0.9036 **	0.7651 **	0.0081	0.2434 .

. p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001. Standard errors are Newey-West HAC (lag = 4).

Asset Class	R ²	Const.		Betas					
		Const.	Const. Ann.	Market	Commodities	Term	Credit	Liquidity	Currency

Regressions with lags (^a, ^b, ^c) report total (Dimson) Beta and joint significance via Wald F-test under HAC vcov.

a) Only includes Market, Liquidity and Currency, 62 obs. b) includes 8 and c) includes 2 lags of Market, Term and Credit.

Table 13: Detailed factor exposure regressions (traditional, commodities, hedge funds)

	S&P 500	IG Bonds	HY Bonds	T-Bonds	Gold Spot	Bitcoin ^a	Comm. Fut.	Man. Fut.	H. Funds	H. Funds (unsm.)
Const.	0.001 (0.001)	-0.000 (0.002)	0.000 (0.001)	0.000 (0.000)	0.018 * (0.007)	0.399 . (0.207)	0.012 ** (0.004)	0.008 * (0.004)	0.005 * (0.002)	0.003 (0.002)
Market	0.942 *** (0.023)	0.011 (0.021)	-0.007 (0.013)	0.007 . (0.004)	-0.026 (0.087)	4.977 * (2.204)	-0.018 (0.066)	-0.015 (0.072)	0.358 *** (0.057)	0.413 *** (0.060)
Commodities	-0.001 (0.013)	-0.002 (0.016)	-0.006 (0.011)	-0.008 . (0.004)	0.308 *** (0.088)		1.022 *** (0.056)	0.121 * (0.057)	0.097 ** (0.032)	0.090 ** (0.033)
Term	0.014 (0.018)	0.713 *** (0.032)	0.436 *** (0.025)	0.701 *** (0.010)	0.602 ** (0.209)		-0.282 ** (0.088)	0.091 (0.127)	0.021 (0.056)	0.055 (0.058)
Credit	0.024 (0.031)	0.446 *** (0.049)	0.988 *** (0.020)	-0.008 (0.008)	0.049 (0.127)		-0.042 (0.138)	-0.183 . (0.094)	0.061 (0.103)	0.044 (0.100)
Liquidity	-0.028 ** (0.010)	0.008 (0.017)	0.000 (0.009)	-0.001 (0.004)	-0.111 (0.073)	-4.816 (3.067)	0.071 . (0.038)	-0.035 (0.036)	0.049 * (0.020)	0.065 ** (0.020)
Currency	-0.016 (0.022)	-0.098 * (0.037)	-0.059 ** (0.018)	-0.005 (0.007)	-0.538 *** (0.138)	-2.821 (4.677)	-0.023 (0.091)	-0.016 (0.138)	0.004 (0.061)	-0.019 (0.055)
Const. Ann. (x4)	0.003	-0.000	0.002	0.000	0.071	1.597	0.046	0.032	0.018	0.012
Ljung-Box p-val. (8 lags)	0.373	0.142	0.054	0.442	0.522	0.881	0.938	0.219	0.001	0.011
Breusch-Pagan p-val.	0.119	0.475	0.006	0.057	0.849	0.302	0.076	0.050	0.160	0.068
White p-val.	0.082	0.000	0.000	0.142	0.479	0.631	0.291	0.139	0.265	0.037
Shapiro p-val.	0.041	0.004	0.034	0.010	0.159	0.000	0.068	0.255	0.000	0.001
Num.Obs.	116	116	116	116	116	62	116	116	116	116
R2	0.988	0.844	0.979	0.992	0.371	0.091	0.846	0.112	0.814	0.822
R2 Adj.	0.988	0.835	0.978	0.992	0.336	0.044	0.837	0.064	0.804	0.812
AIC	-747.3	-686.9	-805.9	-1034.3	-321.6	232.2	-406.6	-412.6	-583.0	-566.9
BIC	-725.2	-664.9	-783.8	-1012.3	-299.5	242.8	-384.6	-390.5	-561.0	-544.9
Log.Lik.	381.633	351.448	410.930	525.141	168.777	-111.100	211.301	214.278	299.513	291.464
RMSE	0.009	0.012	0.007	0.003	0.056	1.452	0.039	0.038	0.018	0.020

. p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Standard errors are Newey-West HAC (lag = 4).

a) Only includes Market, Liquidity and Currency, 62 obs.

Table 14: Detailed factor exposure regressions (private capital)

	Buyout	Buyout (unsm.)	Ven. Cap.	Ven. Cap. (unsm.)	Nasdaq Comp.	Priv. Debt	Priv. Debt (unsm.)	Lev. Loans
Const.	0.017 *** (0.004)	0.012 ** (0.004)	0.012 (0.009)	-0.001 (0.010)	-0.000 (0.005)	0.015 *** (0.002)	0.014 *** (0.003)	0.002 (0.002)
Market	0.356 *** (0.055)	0.492 *** (0.067)	0.865 *** (0.194)	1.340 *** (0.320)	1.368 *** (0.154)	0.005 (0.039)	-0.002 (0.058)	-0.082 . (0.048)
Commodities	0.145 ** (0.053)	0.079 (0.074)	0.256 ** (0.096)	-0.014 (0.156)	-0.156 * (0.063)	0.100 ** (0.036)	0.103 * (0.047)	0.036 (0.024)
Term	-0.071 (0.067)	-0.099 (0.111)	-0.089 (0.205)	-0.371 (0.421)	-0.169 (0.117)	0.143 (0.121)	0.170 (0.157)	0.079 * (0.038)
Credit	-0.028 (0.104)	0.158 (0.124)	-0.811 * (0.359)	-0.701 (0.547)	-0.162 (0.215)	0.478 *** (0.113)	0.561 *** (0.138)	0.698 *** (0.140)
Liquidity	-0.005 (0.034)	0.048 (0.049)	0.022 (0.063)	0.229 (0.151)	0.082 (0.050)	0.029 (0.026)	0.069 * (0.032)	0.027 (0.019)
Currency	-0.158 . (0.081)	-0.295 * (0.132)	0.379 * (0.157)	0.185 (0.200)	0.070 (0.103)	-0.210 * (0.090)	-0.245 * (0.106)	-0.037 (0.062)
Const. Ann. (x4)	0.070	0.047	0.047	-0.006	-0.001	0.060	0.055	0.010
Ljung-Box p-val. (8 lags)	0.000	0.030	0.000	0.052	0.049	0.205	0.052	0.468
Breusch-Pagan p-val.	0.927	0.295	0.046	0.053	0.426	0.173	0.139	0.000
White p-val.	0.583	0.582	0.037	0.070	0.082	0.389	0.408	0.000
Shapiro p-val.	0.238	0.004	0.000	0.000	0.000	0.000	0.000	0.000
Num.Obs.	116	116	116	116	116	116	116	116
R2	0.595	0.628	0.396	0.390	0.854	0.567	0.547	0.796
R2 Adj.	0.573	0.607	0.363	0.357	0.846	0.544	0.522	0.785
AIC	-454.7	-369.1	-284.6	-135.1	-363.3	-475.6	-429.0	-590.2
BIC	-432.7	-347.1	-262.5	-113.1	-341.2	-453.6	-407.0	-568.1
Log.Lik.	235.364	192.569	150.276	75.562	189.633	245.815	222.503	303.082
RMSE	0.032	0.046	0.066	0.126	0.047	0.029	0.036	0.018

. p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Standard errors are Newey-West HAC (lag = 4).

Table 15: Detailed factor exposure regressions (real estate)

	R. Est.	R. Est. (unsm.)	Listed R. Est.	R. Est. ^b	R. Est. (unsm.) ^c
Const.	0.011 . (0.005)	0.004 (0.011)	-0.001 (0.008)	-0.009 (0.006)	-0.012 (0.011)
Market	0.133 (0.091)	0.158 (0.148)	0.498 ** (0.152)	0.003 (0.045)	0.005 (0.081)
Commodities	0.135 * (0.066)	0.299 * (0.141)	0.211 * (0.093)	0.073 * (0.029)	0.175 . (0.094)
Term	-0.048 (0.073)	0.113 (0.266)	0.890 *** (0.196)	-0.057 (0.045)	0.015 (0.192)
Credit	-0.325 . (0.176)	0.139 (0.175)	0.754 *** (0.216)	-0.201 * (0.095)	0.127 (0.138)
Liquidity	0.018 (0.028)	-0.047 (0.085)	-0.152 (0.096)	0.062 * (0.026)	0.008 (0.073)
Currency	0.117 . (0.065)	0.338 (0.206)	0.067 (0.175)	-0.059 (0.052)	0.243 . (0.140)
Market 1 st Lag				-0.073 . (0.037)	0.053 (0.087)
Market 2 nd Lag				0.044 (0.038)	0.391 * (0.163)
Market 3 rd Lag				0.041 (0.043)	
Market 4 th Lag				0.046 (0.028)	
Market 5 th Lag				0.133 ** (0.042)	
Market 6 th Lag				0.093 ** (0.035)	
Market 7 th Lag				0.100 * (0.045)	
Market 8 th Lag				0.041 (0.042)	
Term 1 st Lag				0.164 ** (0.048)	0.629 ** (0.190)
Term 2 nd Lag				0.148 ** (0.046)	0.259 (0.187)
Term 3 rd Lag				0.143 *** (0.041)	
Term 4 th Lag				0.082 . (0.043)	
Term 5 th Lag				0.041 (0.066)	
Term 6 th Lag				0.086 . (0.051)	
Term 7 th Lag				0.142 * (0.056)	
Term 8 th Lag				0.127 ** (0.046)	
Credit 1 st Lag				0.282 * (0.117)	0.793 ** (0.276)
Credit 2 nd Lag				0.098	-0.154

	R. Est.	R. Est. (unsm.)	Listed R. Est.	R. Est. ^b	R. Est. (unsm.) ^c
				(0.068)	(0.267)
Credit 3 rd Lag				0.173 *	
				(0.066)	
Credit 4 th Lag				0.209 ***	
				(0.060)	
Credit 5 th Lag				-0.041	
				(0.061)	
Credit 6 th Lag				0.036	
				(0.060)	
Credit 7 th Lag				-0.044	
				(0.057)	
Credit 8 th Lag				0.060	
				(0.063)	
Const. Ann. (x4)	0.043	0.018	-0.005	-0.035	-0.047
Ljung-Box p-val. (8 lags)	0.000	0.002	0.774	0.000	0.119
Breusch-Pagan p-val.	0.019	0.007	0.001	0.001	0.001
White p-val.	0.002	0.000	0.000	0.015	0.000
Shapiro p-val.	0.000	0.000	0.039	0.001	0.000
Num.Obs.	116	116	116	108	114
R2	0.149	0.164	0.533	0.673	0.451
R2 Adj.	0.102	0.118	0.507	0.545	0.386
AIC	-474.2	-260.3	-278.3	-489.7	-289.7
BIC	-452.2	-238.2	-256.3	-403.9	-251.4
Log.Lik.	245.101	138.131	147.168	276.850	158.872
RMSE	0.029	0.074	0.068	0.019	0.060

. p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Standard errors are Newey-West HAC (lag = 4).

b) Includes 8 and c) includes 2 lags of Market, Term and Credit.

Table 16: Bitcoin quarterly excess returns factor exposure

	Naive	Dummy	Pre 2020	After 2020
Const.	0.399 . (0.207)	0.032 (0.127)	0.497 . (0.262)	0.067 (0.064)
Market	4.977 * (2.204)	4.804 * (2.183)	8.110 * (3.289)	3.737 ** (1.258)
Liquidity	-4.816 (3.067)	-4.757 (2.938)	-18.243 ** (6.328)	-0.214 (0.975)
Currency	-2.821 (4.677)	-3.133 (4.749)	-7.820 (5.571)	4.784 . (2.324)
Pre 2020		0.610 . (0.340)		
Const. Ann. (x4)	1.597	0.129	1.986	0.269
Ljung-Box p-val. (8 lags)	0.881	0.936	0.672	0.688
Breusch-Pagan p-val.	0.302	0.374	0.084	0.362
White p-val.	0.631	0.693	0.006	0.748
Shapiro p-val.	0.000	0.000	0.000	0.002
Num.Obs.	62	62	38	24
R2	0.091	0.129	0.272	0.301
R2 Adj.	0.044	0.068	0.208	0.197
AIC	232.2	231.6	152.9	39.5
BIC	242.8	244.3	161.0	45.4
Log.Lik.	-111.100	-109.776	-71.428	-14.758
RMSE	1.452	1.421	1.585	0.448

. p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Standard errors are Newey-West HAC (lag = 4).

Pre 2020 is a dummy = 1 if date >= 2020-01-01.